Information Systems Frontiers
https://doi.org/10.1007/s10796-024-10487-w

An Economic Framework for Creating AI-Augmented Solutions Across Countries Over Time

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Accepted: 4 April 2024
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Abstract
This paper examines the potential for collaboration between countries with differential resource endowments to advance AI innovation and achieve mutual economic benefits. Our framework juxtaposes economies with a comparative advantage in *AI-capital* and those with a comparative advantage in *tech-labor*, analyzing how these endowments can lead to enhanced comparative advantages over time. Through the application of various production functions and the use of Edgeworth boxes, our analysis reveals that strategic collaboration based on comparative advantage can yield Pareto improvements for both developed and developing countries. Nonetheless, this study also discusses the challenges of uneven benefit distribution, particularly the risk of "brain drain" from developing nations. Contributing to the discourse on the economics of AI and international collaboration, this study highlights the importance of thoughtful strategic planning to promote equitable and sustainable AI development worldwide.

Keywords Artificial Intelligence · Resource Allocation · Production Function · Heterogeneous Country Responses · AI Infrastructure · Tech Labor · Pareto Improvement · Pareto Optimality · International Collaboration

1 Introduction

Artificial intelligence (AI) stands out as one of the most significant general-purpose technologies (GPTs) of our era that holds the potential to be developed as true 'engines of growth' (Bresnahan & Trajtenberg, 1995; Goldfarb et al., 2023). We find ourselves at a pivotal moment in AI development, where the dynamic interplay between innovation in core AI-enabling technologies (such as breakthroughs in algorithms and computer hardware) and innovation in AI application sectors (such as the integration of AI tools to automate and complement the production of goods and services) are becoming increasingly noticeable (Marr, 2023). This synergy is fostering a growing GPT ecosystem, unveiling expansive growth opportunities across various sectors, opening avenues for transformative business processes. Driven by escalating AI demand, AI is poised to set the stage for substantial economic growth.

Considering AI's expanding status as a pivotal GPT, countries around the world are racing to foster environments that are favorable to AI development. It is acknowledged that leadership in AI innovation not only bestows a competitive edge but also facilitates the pivotal role of shaping the trajectory of this transformational technology (Cesareo et al., 2023). Currently, the United States stands as a frontrunner in AI investment and research and development,[1] actively followed by the United Kingdom, China, Israel, Canada, the Netherlands, South Korea, and Germany (Ozkaya & Demirhan, 2023). As countries pour substantial resources into AI development, it becomes evident that achieving AI leadership extends beyond national initiatives; it calls for the establishment of strategic global collaborations that enable the optimized use of differential advantages. A canonical illustration of such a collaboration is the initiative undertaken by NVIDIA of the United States and Reliance Industries of India to establish NVIDIA-powered

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[1] A surge in AI development is predominantly driven by a set of US-based superstar firms (Tambe et al., 2020).

Published online: 24 June 2024

AI supercomputers in India (Singh, 2023). This partnership emphasizes the strategic alignment of the United States's technological expertise and India's vast pool of tech skill talent and sets a precedent for global AI development cooperation.

In this paper, we propose an economic framework designed to explore how countries, each with differential comparative advantages, can progressively collaborate to foster innovation in AI. Our primary goal is to conceptualize the AI innovation process from a country-level perspective to assess potential outcomes of such partnerships. To do so, we formally outline the following research questions:

- How do countries with differential resource endowments contribute to the development of AI technologies over time, and what are the implications for global economic development?
- In what ways can international collaborations in AI innovation lead to shared prosperity across nations?

In addressing these questions, we propose a dyadic structure that encompasses two distinct types of economies, characterized by comparative advantages in either *AI-capital* or *tech-labor*. Within this framework, *AI-capital* encompasses both tangible and intangible information technology (IT) assets. Tangible IT assets include various types of hardware, cloud computing infrastructures, GPUs, data centers, and supercomputers designed for extensive computations. Intangible assets include various software solutions, data repositories, and the accumulated data resources critical for machine learning (ML) algorithm development. The *tech-labor* represents human resources specialized in the development, management, and maintenance of AI/ML-related solutions. This includes skills in data management, programming, and algorithm development; skills related to data analytics and structured decision-making apparatuses; and the human expertise essential for creating AI innovation. To depict a long-term collaboration between economies distinguished by their comparative resource advantages, our dyadic scenarios involve developed and developing economies, such as the United States, with its significant investment in *AI-capital* and higher *tech-labor* costs, and India, known for its *tech-labor* abundance and lower costs.

Note that this structure builds on the classical economic model's foundational assumption of two production factors, capital and labor, yet incorporates the concept of digital intangible capital (Tambe et al., 2020). Digital intangible capital underscores the necessity of investment in digital complementarities to unlock the potential of GPTs like AI. These intangible assets often represent a substantial portion of the total costs associated with GPT deployment. Moreover, due to the scarcity and considerable amount of time required for development and accumulation, digital intangible capital highlights the importance of strategic investment and potential collaboration in AI-related digital infrastructure. Previous research has primarily quantified these digital intangible IT assets through proxies including IT employment (Tambe & Hitt, 2012), IT investment (Bresnahan et al., 2002), or a combination of computer capital and information systems labor (Hitt & Brynjolfsson, 1996). *AI-capital* and *tech-labor* in our context are essential resources to drive AI innovation and create new AI-augmented business models that were not possible with conventional IT technologies. Unlike existing studies, our framework shifts the focus from quantification to the conceptual integration of these elements into the foundational economic principles of capital and labor. By doing so, we explicitly recognize the role of and incorporate digital intangible capital in our framework.

In our framework, *AI-capital* and *tech-labor* inputs work together to foster a singular output, q, which we define as the *AI-augmented solutions*. This term broadly spans across a variety of products, services, and applications that are augmented by AI technologies, resulting in improved functionality and broader capabilities. This construct is marked by two distinct characteristics. First, while related to digitalization, *AI-augmented solutions* diverge from traditional digitization outputs by imitating human intelligence, necessitating both *AI-capital* and *tech-labor* for development. Second, our output measure focuses on AI innovation, leaning more toward the *creation* of AI rather than its *usage* to produce goods and services. We highlight the nuanced distinction between *creating* innovations for a GPT at the application sector level and simply *utilizing* these technologies. Successful dissemination of GPTs, such as AI, requires application sector innovations (Bresnahan & Trajtenberg, 1995; Goldfarb et al., 2023), and thus, *AI-capital* and *tech-labor* are essential for this process.

To illustrate the dynamics of *AI-augmented solution* production, our theoretical framework incorporates three classical production functions—Cobb–Douglas, Linear, and Leontief—comprehensively encompassing all foundational assumptions of production. This exhaustive approach ensures our analysis captures the full spectrum of potential production dynamics in the context of the rapidly developing, yet still nascent, domain of *AI-augmented solution* development. Drawing from foundational works of technological lock-in (Arthur, 1989), path dependency (Arthur, 1994), and skill-biased technological change as highlighted by Acemoglu (2002), we assume that countries will advance along the technological advancement trajectories that align with their inherent comparative advantages. This suggests that countries rich in *tech-labor* will continue to develop labor expertise, while countries with an abundance of *AI-capital* will focus on enhancing technological infrastructure. Additionally, countries advanced in both domains are

expected to progress with advancements maintaining the same ratio between the two, adhering to a Hicks-neutral technological progress assumption. This leads to a pivotal assumption in our framework: AI technology induces "differential shifts" in the isoquants, signifying varying development trajectories for countries based on their comparative advantages. Technological progress not only shifts the factor price frontier of *AI-augmented solution* production outward (Fig. 8), but also results in a "differential tilt" of the isoquant shift toward the resource that each economy has a comparative advantage in (Figs. 1 and 2). The dynamics between countries are visually depicted through the Edgeworth box diagram, illustrating the potential for optimized production strategies through collaborative efforts in global AI development.

The central theme emerging from our analytical model is that the classical Ricardian Model of Comparative Advantage fundamentally applies to country-level AI collaboration. Regardless of the production function employed, when economies leverage their comparative advantages through mutual engagement, both experience Pareto improvements. However, when trade is initiated unilaterally, only the initiating economy achieves Pareto improvements, while the output for the non-initiating economy remains unchanged. It is critical to note that Pareto optimality does not guarantee equitable resource distribution; many distributions can achieve Pareto optimality, including those that might appear inequitable in the long run. Following the results from our analytical model and data simulation, we illustrate scenarios where developing economies, through trade with *AI-capital*-intensive economies, can capitalize on foreign investments. Conversely, scenarios like the "brain drain" of *tech-labor* illustrate the complexities and challenges in ensuring equitable benefits from AI advancements across different countries. Potentially, these outcomes could lead to the emergence of "superstar countries" that monopolize the benefits of AI development (Korinek & Stiglitz, 2021).

This study contributes to the literature on the economics of AI, focusing on the long-term implications of international collaboration on AI innovation. While research on the economics of AI is rapidly expanding, the study of AI and international collaboration remains significantly unexplored. Trefler and Sun (2022) highlight this significant research gap, stating, "Artificial Intelligence is a powerful new technology that will likely have large impacts on […] international trade flows. Yet almost nothing is known empirically about this." Few researchers explores AI's influence on international trade, laying the theoretical foundation for understanding how AI might reshape trade dynamics (Goldfarb & Trefler, 2018). Trefler and Sun (2022) offer an empirical perspective, examining concrete examples of AI's effect on trade patterns. Our work seeks to build upon these foundational insights by exploring the long-term dynamics of country-level collaboration in AI development, to shed light on how such cooperation might yield mutual benefits and shape the future of collaborative global AI development.

Our framework introduces a strategic lens for evaluating the multifaceted economic impacts of AI on international development and collaboration. It underlines the prospect of achieving mutually beneficial Pareto-improved economic situations (Rauch & Casella, 2003) through well-coordinated exchanges, while also addressing the potential pitfalls, such as the depletion of intellectual resources in less developed countries. By framing complex global economic interactions into a simplified dyadic model that contrasts *AI-capital* with *tech-labor* advantages, we provide stakeholders with a more nuanced perspective of AI's potential impact on future economic landscapes. This approach highlights effective strategic approaches for international collaboration, urging caution against imbalanced exchanges that could lead to potential "brain drain" in developing countries, thus offering insights that can inform policy formulation and strategic decision-making in the AI era.

The rest of the paper is outlined as follows: The next section provides a concise overview of the existing literature on GPTs, the conceptualization of AI as an automation technology, the characteristics of digital intangible assets, and the dynamics of AI in an international trade context. It addresses key questions regarding the augmentation of innovation processes through AI, specifically focusing on how AI, with its distinctive capabilities, differs from previous technological advancements in automation and digital transformation in terms of economic impacts and innovation dynamics. Building on these foundational concepts, Section 3 presents our framework, distinguishing between economies intensive in *AI-capital* and *tech-labor*. It defines key terms and lays the foundation of our dyadic model, emphasizing the specifics of *AI-augmented solutions*, the three production functions, and visual depictions through Edgeworth box diagrams. In addition, key assumptions, including the assumption of a "tilting slope," are elaborated in the development paths of economies based on their comparative advantages. Section 4 examines the economic potential of various collaborative resource allocation approaches, focusing on different production function assumptions and strategic initiations. Section 5 summarizes the findings and adds discussions that connect our analysis with broader economic perspectives. We conclude in Section 6 with suggestions for future research directions in this rapidly evolving field.

2 Literature Overview

The rapid growth of AI development has captured significant scholarly interest, with leading economists exploring its potential and challenges. Considering the multifaceted

nature of AI as a GPT, scholars have conducted a wide array of research. Below, we outline a few key studies within the economics literature that have profoundly shaped our comprehension of AI's diverse impacts and discuss how these studies shaped our conceptual framework.

- As foundational works, Brynjolfsson and McAfee (2014) characterized the upcoming AI era as the "second machine age," creating widespread interest. Varian (2018) outlined the complex impact of AI on organizations from an economic perspective, emphasizing the challenges related to the scarcity of essential resources such as data, algorithms, and hardware.
- Significant attention in the field has been directed toward understanding AI's influence on automation, job replacement, and unemployment (Acemoglu & Restrepo, 2018; Acemoglu et al., 2022; Autor & Dorn, 2013; Babina et al., 2022; Brynjolfsson & Mitchell, 2017; Felten et al., 2021; Frey & Osborne, 2017; Webb, 2019).
- A set of studies underscored the importance of firm-level investments in IT, ML, and AI for creating intangible capital, examining how such investments correlate with improvements in innovation and productivity (Bresnahan et al., 2002; Brynjolfsson et al., 2017; Hitt & Brynjolfsson, 1996; Tambe & Hitt, 2012; Tambe et al., 2020; Wu et al., 2020).
- Exploration of AI's impact on economics extends into several niche areas that provide detailed insights into specific aspects of its influence: policy-related studies by Aghion et al. (2018) examine economic growth and societal integration facilitated by AI; Eloundou et al. (2023) highlight the role of Large Language Models as pivotal GPTs; Acemoglu (2021) scrutinizes the potential for AI to exacerbate social inequalities; and Beraja et al. (2023) analyze the impact of AI's deployment within autocratic political systems.
- In the domain of AI and international trade, Goldfarb and Trefler (2018) conceptualize AI's potential reshaping of international trade dynamics, and Trefler and Sun (2022) empirically assess AI's international impacts with mobile application downloads.

2.1 Theories of Comparative Advantages and General-Purpose Technologies

The intersection of the theory of comparative advantage and AI technology provides a valuable perspective on the international trade dynamics of AI. The classical theory, originally posited by David Ricardo, suggests that countries benefit from international trade by specializing in the production of goods for which they hold a relative efficiency advantage. This specialization allows countries to engage in mutually beneficial trade, maximizing economic welfare by focusing on their comparative advantages. The principle of comparative advantage centers on the idea of opportunity cost and applies even in cases where one country is more efficient in the production of all goods compared to another country, and hence, emphasizes relative efficiency over absolute superiority. If we apply the theory of comparative advantage to our dyadic framework, despite the United States possessing superior *AI-capital* and *tech-labor* on a global scale, the United States can still benefit from trading with India, where *tech-labor* costs are relatively lower.

AI's designation as a pivotal GPT suggests how the classical comparative advantage theories can be applied and extended. Unlike conventional goods, the maturation of GPTs necessitates advancement not just in the core technology but also across sector-specific applications. This requirement establishes a feedback loop where innovation in one domain spills over to another, collectively accelerating the technology's evolution and integration into various sectors. Hence, such dynamics underscore the importance of international collaboration among nations possessing multifaceted comparative strengths (Bresnahan & Trajtenberg, 1995), highlighting the mutual benefits of leveraging distinct capabilities to drive collective progress in AI innovation. Recent studies have identified subsectors of AI, including ML and generative AI, as pivotal GPTs due to their profound economic impact and versatility across sectors (Eloundou et al., 2023; Goldfarb et al., 2023). This broad applicability of AI as a GPT underscores the need for multifaceted *AI-augmented solutions*, which can be strategically realized through the integration of various forms of *AI-capital* and *tech-labor*.

We revisit the partnership between NVIDIA and Reliance Industries as a pivotal case in strategic international collaboration for AI development. This alliance aims to establish a supercomputing infrastructure in India, specifically to generate foundational large language models that leverage the country's extensive linguistic diversity. This project taps into India's abundant *tech-labor* as well as India's massive linguistic data. Note that linguistic data serves as a complementary asset, a crucial form of *AI-capital*, in this specific context. By merging NVIDIA's advanced technological infrastructure with India's rich linguistic resources and skilled tech workforce, this venture highlights the importance of strategic partnerships in driving forward the innovation of AI technologies. This case demonstrates the power of collaboration to harness unique capabilities from each partner, fostering AI innovation that meets a wide array of multifaceted requirements.

2.2 Digital Intangible Capital

Building on the characteristics of AI as a GPT, we draw from the IT productivity literature to highlight the need of complementary assets for AI innovation. The IT productivity

literature shows that the mere acquisition of IT solutions falls short of optimizing business processes. For IT to reach its full effectiveness, complementary investments in IT skills, organizational structure, data infrastructure, a hardware and software ecosystem, and more are required (Hitt & Brynjolfsson, 1996; Tambe & Hitt, 2012). As IT evolves into more complex AI systems, this principle becomes increasingly pertinent. Brynjolfsson et al. (2017) highlight that with AI, the world appears to be witnessing a redux of the IT productivity paradox,[2] indicating that the challenges observed during early IT integration phases will resurface in the AI context. Furthermore, Wu et al. (2020) demonstrate that data analytics are complementary to IT innovation, required for productivity gains. This concept is further analyzed by Tambe et al. (2020) under the construct of digital intangible capital, highlighting the critical role complementary assets play in driving productivity and innovation in AI.

Digital intangible capital shares similarities and differences with conventional IT complementary. Both share foundational IT requirements and are typically measured through conventional IT investment proxies (Bresnahan et al., 2002; Hitt & Brynjolfsson, 1996; Tambe & Hitt, 2012). However, AI complementary exhibits distinct characteristics, notably the need for extensive, sophisticated datasets that drive economies of scale and network externalities (Goldfarb & Trefler, 2018). This cycle, where advanced data enhances algorithms, improves platform capabilities, attracts more users, and in turn generates more data, is demonstrated by Google's pursuit of quality data through its data acquisition partnership with Reddit (Isaac & Hirsch, 2024). Additionally, AI uses economies of scope, with datasets serving multiple applications, thereby extending the utility of each data point. Knowledge externalities in AI are characterized by a global dissemination of knowledge, yet with a notable concentration of talent in specific regions, which emphasizes the need for strategic collaborations. In our framework, we depict how digital intangible capital, critical for AI innovation, requires collaborative global efforts to develop, highlighting the imperative of combining *AI-capital* and *tech-labor* in the global landscape.

2.3 Automation and Task Model

Drawing from the literature on AI and automation, our framework employs the three foundational production functions. Note that the literature on AI and automation provides insights into the changing dynamics of the labor market. Frey and Osborne's early work (2017) marked a pivotal moment, sparking widespread scholarly interest and paving the way for subsequent studies in this field. Another significant work in this field has been the focus on the task model (Acemoglu & Autor, 2011; Acemoglu et al., 2022), which shifted the discussion away from the broad scope of occupational automation to the detailed scope of task automation. Through this lens, tasks can be categorized into three groups based on automation potential: those highly amenable to automation, those that present challenges to automation due to technical or social reasons, and a wide spectrum of tasks that lie in between.

The three foundational production functions we use—Cobb–Douglas, Linear, and Leontief—resonate with the task model's conceptualization of AI and automation. The task model highlights how AI technology can both substitute for and augment (i.e., complement) human capabilities, thereby transforming business processes. Our choice of production functions is informed by the task model's categorization of tasks based on susceptibility to automation. For instance, the Linear production function represents tasks that are completely automatable by AI, symbolizing activities where technology can fully take over without human intervention. A historic occupational example includes the role of telephone switchboard operators, whose primary function was to connect calls manually. With advancements in technology, the tasks became fully automated, exemplifying the kind of production tasks captured by the Linear production function. This showcases the scenario for *AI-capital* and *tech-labor* to perfectly substitute for each other under certain conditions, and our framework outlines strategic considerations essential for optimizing production across diverse economic contexts.

The Cobb–Douglas production function applies to the majority of tasks that are partially automatable by AI, such as the case with semi-autonomous vehicles. In these instances, the process of driving is partially automated, but not to the extent of full autonomy. It's important to note that the chosen production function is adaptable; as technology advances—potentially to the point of reaching singularity (Korinek & Stiglitz, 2021) and achieving complete autonomy in driving (referred to as level 5 autonomy)—the production function may shift to a Linear model to reflect full automation. Meanwhile, the Leontief production function captures scenarios in which human oversight or specific computational tasks are indispensable. This includes situations that demand the discernment of a medical professional in surgical operations. Collectively, these examples highlight the diversity of tasks and the nuanced application of AI in automating or augmenting human tasks, underscoring the need for a flexible approach in conceptualizing how *AI-capital* and *tech-labor* interact within our economic model.

We emphasize that our application of production functions is deliberately centered on AI innovation, through examining AI's role in task automation and its contribution

[2] The IT productivity paradox refers to academic findings in the late 1980s and early 1990s that showed significant investments in IT do not always lead to proportionate increases in productivity.

to technological advancement. Contrary to the predominant focus on AI's automation and enhancement of general tasks (i.e., *using* AI),[3] our framework applies traditional production functions to specifically encompass the development of AI innovations (i.e., *creating* AI). Consider how the advancement of ChatGPT and large language models represents a paradigm shift in AI innovation. Tasks that previously required extensive *tech-labor* are increasingly being substituted by sophisticated *AI-capital*. For instance, where a team of researchers might have been needed to parse through data, develop algorithms, and refine outputs, large language models can now perform similar functions with greater speed and efficiency. This shift exemplifies the substitution of *tech-labor* with *AI-capital* and demonstrates the task model in an AI innovation context.

2.4 AI and International Trade

Literature that specifically addresses AI and international trade remains limited. A few studies are indirectly related to our research. Brynjolfsson et al. (2018) provide empirical evidence on how AI, specifically through eBay's implementation of machine translation technology, can significantly enhance trade activities by boosting exports. This study underscores the direct impact of AI technologies on facilitating international commerce. Similarly, Stapleton and Webb (2020) examine the effects of robotics adoption on the international dynamics of offshoring and reshoring, highlighting the role of automation in shaping global trade patterns. Yet these explorations differ from our conceptual framework. We posit broad AI as a GPT, spanning beyond robotics, necessitating the extensive integration and development of digital intangible assets.

The works of Goldfarb and Trefler (2018) and Trefler and Sun (2022) are the closest to our work. The studies stand out in their detailed exploration of how AI technologies influence global trade dynamics. In our discussion on digital intangible capital, Goldfarb and Trefler (2018) provide a foundational framework on AI's role in international trade, emphasizing economies of scale from data, the scope from diverse applications, and knowledge externalities. The study illustrates how data accumulation and the strategic use of *AI-capital* can enhance global trade dynamics, particularly enabling developing countries to leverage comparative advantages for a stronger position in the international market. Trefler and Sun (2022) complement this study by offering empirical insights into AI's tangible impacts on trade flows, emphasizing the role of AI in altering comparative advantages and trade strategies. Together, these studies form a pivotal basis for our study, guiding our analysis of AI's role in fostering international collaboration and reshaping the landscape of global commerce.

3 Model

In this section, we draw from the extant literature reviewed to formalize the assumptions and foundational elements of our analytical framework. A dyadic model with differential resource endowments is presented, focusing on the collaborative dynamics between economies to foster the development of a singular *AI-augmented solution*. By delineating key terminologies, we lay the groundwork for an in-depth exploration of the three classical production functions—Cobb–Douglas, Linear, and Leontief—and their representation through Edgeworth box diagrams. Central to our analysis is the exploration of the "tilting slope" assumption, a concept that depicts divergent developmental trajectories for economies based on their comparative advantages in *AI-capital* and *tech-labor*. This assumption is pivotal for understanding how distinct economies can strategically employ comparative advantages to catalyze innovation, accelerate economic growth, and position themselves within the rapidly evolving global AI landscape.

In economic theory, the production functions are traditionally broken down into "capital" and "labor" as the primary factors of production. The term "capital" covers a broad range of elements, including human-made resources such as machinery and tools (Varian, 1992), both public and private infrastructures essential for producing goods and services (Turnovsky, 1997), the services derived from various capital goods (Solow, 1955), and the monetary values assigned to these capital goods, encompassing interest and related rates (Robinson, 1953). In addition to these classical components, our model integrates the concept of intangible digital capital (Tambe et al., 2020), highlighting the critical role of intangible asset investments in harnessing the full potential of a GPT like AI. These intangible assets often compose a significant share of the overall costs involved in deploying GPTs. In our research context, *AI-capital* is viewed as a catalyst for innovation and the creation of novel business models (i.e., AI products and AI services), enabling possibilities that extend beyond the capabilities of traditional IT technologies.

We conceptualized *AI-capital* as a category that includes both tangible and intangible assets within the scope of IT. Furthermore, this category of intangible assets extends to include legal frameworks and property rights structures, thereby facilitating the acquisition of data from users and

[3] To concentrate on AI *usage*, the existing literature often excludes task activities that pertain to the North American Industry Classification System (NAICS) sectors 51 (Information) and 54 (Professional, Scientific, and Technical Services) as these sectors are generally AI *creating* industry sectors. When we focus on *using* AI, the AI impact on tasks can be viewed in line with conventional technology where it is conceptualized as a factor that substitutes and/or complements tasks. We extend the concept of AI's impact on tasks to specifically AI innovation tasks, drawing from Aghion et al. (2018). See Babina et al. (2022) and Acemoglu et al. (2022) for further details.

citizens by *AI-capital* entities. Some academics propose viewing the data, infrastructure, social-legal structure, and mechanized labor (i.e., machine labor) used by organizations or governmental bodies as a foundational element of production within the platform economy context (Wagner, 2020). Conversely, the concept of labor might be interpreted either as the aggregate of labor inputs (Varian, 1992) or as a unit of uniform labor hours (Solow, 1955). In the context of our study, *tech-labor* denotes the workforce specialized in the creation, oversight, and upkeep of AI/ML technologies. These scenarios further incorporate variations in *AI-capital* intensity and *tech-labor* resource allocation. We examine the above research questions through the lens of several economic concepts, specifically (1) the production function, (2) the marginal rate of technology substitutes and the elasticity of substitution, and (3) the Edgeworth box.

3.1 Production Function

In this study, the production function can be taken as the technological apparatus used by aggregated entities within a given country to transform various permutations of inputs into corresponding outputs. In the context of this analysis, inputs and outputs are regarded as flow terms, denoting that a quantified quantum of inputs within a stipulated timeframe is engaged to yield a commensurate quantity of outputs within the same temporal unit (Mas-Colell et al., 1995). Consequently, in our analytical framework, we explicitly incorporate the temporal dimension encompassing specific input–output pairings.

For analytical coherence, we adopt a simplified dyadic framework encompassing two economies, where the inputs—*tech-labor* and *AI-capital*—are harnessed to yield a singular output, q, the *AI-augmented solution* in each economy. Moreover, we posit that the aggregate quantities of *tech-labor* and *AI-capital* in the two economies remain unaltered over the course of our analytical interval. Nonetheless, propelled by the escalating demand for AI, entities within each nation may adjust their input permutations to yield desired quantities of a single *AI-augmented solution*, aligned with their respective production functions, over the course of our analytical interval.

Our study integrates three distinct production functions as integral components of our analytical framework: the (1) Cobb–Douglas, (2) Linear, and (3) Leontief production functions, where $\beta = (1 - \alpha)$ and $0 \leq \alpha, \beta \leq 1$, and K is *AI-capital*, L is *tech-labor*, and q is the quantity of a single *AI-augmented solution* created by the combination of inputs.[4]

$$\overline{q}_t = tK^{\alpha}L^{\beta} \tag{1}$$

Partial Factor Complementary Production Function

$$\overline{q}_t = t(\frac{K}{a} + \frac{L}{\beta}) \tag{2}$$

Perfect Factor Substitution Production Function

$$\overline{q}_t = t[\min\left(\frac{K}{a}, \frac{L}{\beta}\right)] \tag{3}$$

Perfect Factor Complementary Production Function

3.2 Marginal Rate of Technical Substitution and Elasticity of Substitution

The *marginal rate of technical substitution* (*MRTS*) in our study context denotes the mechanism by which each country modulates the quantity of *AI-capital* (K) while holding the output (volume of the *AI-augmented solution*) constant in response to a marginal alteration in *tech-labor* inputs (L). Since we assume "holding the output constant," we can use the total differential of the product function (f) to grasp the meaning of *MRTS* (see Eqs. 4 to 6).

$$d\overline{q} = \frac{\partial f}{\partial L}dL + \frac{\partial f}{\partial K}dK \tag{4}$$

Since output remains the same,

$$0 = \frac{\partial f}{\partial L}dL + \frac{\partial f}{\partial K}dK \tag{5}$$

$$\therefore \frac{dK}{dL} = -\frac{\partial f/\partial L}{\partial f/\partial K} \tag{6}$$

As Eq. (6) shows, the *MRTS* of *tech-labor* for *AI-capital* is the slope of the production functions. Therefore, in the context of our study, we can express the *MRTS* as $MRTS_{LK}$, and it is the slope of each isoquant (production function): $-\frac{\partial f/\partial L}{\partial f/\partial K}$.

While $MRTS_{LK}$ is the slope of an isoquant, the elasticity of substitution (σ) is the ratio of percentage change in resources with the percentage change in *MRTS*. Equation (7) represents the elasticity of substitution.

$$\sigma = \frac{\Delta(\frac{K}{L})/\frac{K}{L}}{\Delta(MRTS_{LK})/MRTS_{LK}} \tag{7}$$

Applying logarithmic derivatives,

$$\therefore \sigma = \frac{d\ln(K/L)}{d\ln|MRTS_{LK}|} \tag{8}$$

[4] Graphical representations of the production functions and supplementary notes on the isoquants are presented in the appendix for further reference. Each isoquant shifts upwards and to the right along a ray originating from the origin.

Based on Eq. (8), therefore, the Cobb–Douglas production function (Eq. 1) has an elasticity of substitution of one (1), assuming that *tech-labor* and *AI-capital* can be substituted at a constant rate, while the Linear production function (Eq. 2) has an infinite (∞) elasticity of substitution, representing perfect *tech-labor* to *AI-capital* substitution. Finally, the elasticity of substitution of the Leontief production function (Eq. 3) is zero (0), demonstrating complementary relationships between the two factors.

The elasticity of substitution for the Cobb–Douglas production function of one implies that alterations in the input of production factors are relatively minor. In simpler terms, when there is a modification in the input of one production factor, the corresponding change in the other input is not substantial. For the Leontief production function, the elasticity of substitution is zero; hence, within the Leontief production function, it is essential to combine the two production factors at a fixed ratio. Regardless of the extent to which the input of one production factor increases, its marginal production remains zero unless it is used in conjunction with the other production factor. On the contrasting end of the spectrum from the Leontief production function lies the Linear production function with its infinite elasticity of substitution. This value implies that no difficulties exist in substituting one input with another within the framework of the Linear production function.

3.3 Edgeworth Box

In the context of a two-agent, two-endowment, and often two-product scenario, a pragmatic method for visualizing allocations of resources, preferences, and endowments in a two-dimensional configuration is the Edgeworth box. The generality of the term "good" (or "product") encompasses a wide spectrum, ranging from temporal and spatial distinctions to variations in the world state (Varian, 1992). Accordingly, under the purview of the Armington assumption, whereby agents perceive identical goods (or identical products) in differing locales as distinct entities (Blonigen & Wilson, 1999; Davis, 1995; Markusen & Wigle, 1990), the framework is equipped to accommodate the portrayal of two discrete entities—developed and developing countries—operating as agents. Herein, the output of *AI-augmented solutions* generated by each of these agents assumes prominence as the focal commodities. Consequently, the Edgeworth box adeptly demonstrates the allocation of endowments by two distinct agents toward the production of a singular product—in our research context, the *AI-augmented solution*.

In the subsequent sections of our analysis, our examination will explore the intricacies of international cooperation concerning resource allocation aimed at facilitating the development of a significant quantity of a single *AI-augmented solution*. To understand the complexities of such collaboration, we will leverage the Edgeworth box framework instead of other trade models like the *Heckscher-Ohlin (H–O) model*. Most trade models, including the *H–O model*, rely on assumptions centered around the difference between two distinct products using each of the countries' abundant factors. For instance, the *H–O model* analyzes exports and imports of a labor-intensive good (like textiles) in a developing country and a capital-intensive good (like machinery) in a developed country. While our study acknowledges varying factor endowments of *tech-labor* and *AI-capital* in different countries, it has a unique focus on the production and collaboration of a large volume of a singular *AI-augmented solution*. Therefore, this setting of a singular product makes other trade models like the *H–O model* less applicable to our research situation.

Consequently, to effectively illustrate the impact of international collaboration on the allocation of resources and the resultant volume of a singular *AI-augmented solution*, the Edgeworth box emerges as the most appropriate analytical tool. This framework facilitates a nuanced exploration of how differential endowments (resources) across nations can converge through collaboration to enhance the production of an *AI-augmented solution*, providing a clear depiction of the interrelationships between varying resource allocations and production outcomes.

3.4 "Differential Shifts" of the Isoquants

The foundational assumption underlying our analysis integrates classical economic theories with contemporary insights on technological evolution, focusing on the nuanced interplay between production functions, the *MRTS*, and the elasticity of substitution. By adopting a framework that highlights the roles of *tech-labor* and *AI-capital* in the production of an *AI-augmented solution*, we draw upon the seminal theories of technological lock-in and path dependency as articulated by Arthur (1989, 1994). This perspective is particularly relevant when considering the differential advantages of nations in *tech-labor* and *AI-capital*. For instance, countries historically focused on skill-intensive technological advancements are likely to see sustained growth in sectors that enhance these capabilities, in line with the principles of skill-biased technological change (Acemoglu, 2002). Such focus on skill-intensive technologies often leads to the development of education and training systems that are specifically designed to reinforce these skills, which overtime creates path dependency, making it challenging and costly to deviate from established skill trajectories (Arthur, 1994). Conversely, nations that have invested

heavily in *AI-capital* are positioned to further enhance their capacities in this domain, potentially leading to a junction in developmental trajectories. Thus, our foundational assumption posits that the interplay between production functions and historical economic patterns shapes the landscape of international collaboration in AI, highlighting the strategic importance of leveraging complementary strengths across nations to optimize the global production of *AI-augmented solutions*. This approach not only acknowledges the legacy of past economic and technological choices but also frames the path forward in the AI era, emphasizing the critical role of strategic international collaboration in navigating the complexities of technological evolution and economic development.

Central to our analysis is the assertion that advancements in AI technology induce "differential shifts" in the production isoquants. Unlike traditional models within the study of production functions, where the *MRTS* remains constant at any specific point in time—a concept traditionally depicted in the academic literature and illustrated by our reference Fig. 8—our approach introduces a dynamic modification. We propose a model where rays, extending from the origin, adjust dynamically to represent shifts in resource competitiveness. This adjustment leads to variations in the *MRTS* at distinct temporal points, a concept illustrated in Figs. 1 and 2 of our analysis. This critical adjustment in our analytical model brings to the fore the dynamic nature of resource allocation and use amid the evolution of AI technology. The consequential shifts and their implications for production functions will be further elaborated upon in Section 4, highlighting the significant impact of these technological advancements on economic models and theories.

4 Analysis

The focal objective of this study is to provide insight into how heterogeneous economies, characterized by differential comparative advantages regarding resource endowments, might respond to creating a large volume of a single *AI-augmented solution* over time. To address this research question, we construct a comparative framework involving two markedly distinct economies, each characterized by diverse levels of *tech-labor* and *AI-capital* resource intensity. For the purposes of analytical rigor and the facilitation of nuanced result interpretation, this study integrates and modifies classical economic constraints. In particular, the constraints underscore the dynamic nature of comparative advantages associated with resource endowments in each economy, positing that these advantages evolve over time. Therefore, the rays emanating from the origin are dynamically recalibrated to mirror the competitiveness of resource endowments in different countries. Simultaneously, this study asserts that the cumulative quantities of *tech-labor* and *AI-capital* resources, when considered across both economies under examination, will maintain a state of constancy throughout the duration of the analysis. This methodology enables a comprehensive exploration of the interplay between static resource allocation and the dynamic evolution of comparative advantages in the context of *AI-augmented solution* development.

In conjunction with these foundational assumptions drawn from classical economics, a fundamental proposition is introduced, asserting the constancy of the production function for generating identical products across heterogeneous economies, akin to the conventional paradigms found in studies of international trade and economic growth (Chipman, 1986, 1988; Davis, 1995). In contrast, our investigation considers a spectrum of production functions pertaining to *AI-augmented solution* development, acknowledging the prevailing lack of consensus or empirical validation concerning the specific production functions that govern the creation of a large volume of a single *AI-augmented solution*.

Within the context of these heterogeneous economies characterized by a duality of resources (*tech-labor* and *AI-capital*), we anticipate that each economy will exhibit distinctive responses to the fluctuating relevant prices associated with these two factors over time. Consequently, we foresee a relative depreciation of one factor (*tech-labor* or *AI-capital*) over time. As a result, in the pursuit of enhancing the volume of a single *AI-augmented solution*, each country is inclined to employ a greater quantity ($r\%$) of the resource in which it holds a comparative advantage. This means that each country might engage $r\%$ ($0 \leq r \leq 1$) more of a particular resource (either *tech-labor* or *AI-capital*) than it would in the absence of this resource price change, taking advantage of the resource's reduced cost. This corollary assumes paramount importance and is simultaneously well-founded, as it acknowledges the potential existence of spillover effects (Baldwin et al., 2005; Chang & Gurbaxani, 2011), capital deepening effects (Glover & Short, 2020; Growiec et al., 2018; Kumar & Russell, 2002), and/or labor-abundant effects (Mamun et al., 2015; Marjit et al., 2019) that could collectively engender an amplified volume of the *AI-augmented solution* over time.

An example of the principle to "employ a greater quantity (r%) of the resource in which it holds a comparative advantage" can be found within the legal frameworks that streamline the process of data acquisition from users and citizens. As an integral component of *AI-capital*'s intangible assets, the legal infrastructure significantly enhances the efficiency of aggregating a substantial and reliable dataset over time. Consequently, the amassed

data not only serves as a cost-effective resource but also potentially catalyzes the emergence of innovative spillover concepts for the development of *AI-augmented solutions*. This paradigm underscores the strategic importance of leveraging comparative advantages in legal and regulatory frameworks to foster the growth of a data-driven ecosystem conducive to AI innovation.

Furthermore, in scenarios where a nation possesses a substantial number of *tech-labor* individuals who have accrued expertise in AI/ML over a prolonged period, the remuneration associated with such labor forces is expected to depreciate. This depreciation in labor costs is conducive to the cost-effective development of *AI-augmented solutions*. This economic phenomenon underscores the interplay between labor market dynamics and technological advancements, illustrating how an increase in the availability of specialized skills can lead to a reduction in wages due to supply surpassing demand. Consequently, this adjustment in the cost structure facilitates a more economically viable environment for the innovation and deployment of a large volume of a single *AI-augmented solution*.

Over the course of time, as the popularity of AI continues to rise, both economies may exhibit a common proclivity to augment their aggregate volume of the *AI-augmented solution*. Consequently, each country endeavors to judiciously select an optimal amalgamation of factor inputs to enhance its production capacity. Since the relevant price of one resource becomes more cost-effective than the other, countries will use more resource ($r\%$) of either *tech-labor* or *AI-capital*. Then, the slope ($MRTS_{LK}$) of the isoquant with the optimal combination of factors in the given time changes compared to the preceding stage (see Figs.1 and 2).

Our analysis indicates that in the initial stages, the two economies refrain from resource trade, despite possessing distinct strengths in their respective resource endowments. This restraint stems from the fact that both economies harbor a surplus of resources adequate to facilitate an increase in the volume of *AI-augmented solutions*. Nevertheless, as time progresses, a shift transpires wherein the two economies begin to explore resource exchange avenues, leveraging their comparative advantages to enhance the *AI-augmented solution* outcomes. Subsequently, the initiation of trade between the two economies materializes solely under conditions wherein the proposed trade, manifesting as alterations in factor allocations, confers benefits to at least one economy without causing harm to the other (Pareto improvement). Ultimately, trade materializes within a framework that attains Pareto optimality, denoting a state wherein no further enhancements can be achieved for either country through the reallocation of resources without rendering one or both countries worse off. This signifies a scenario wherein trade between the economies is structured to the point where optimal outcomes are achieved, aligning with the principles of Pareto efficiency.

4.1 Production Functions in Developed and Developing Economies

Our theoretical framework incorporates three production functions—Cobb–Douglas, Linear, and Leontief—comprehensively encompassing all foundational assumptions of production. This exhaustive approach ensures our analysis captures the full spectrum of potential production dynamics in the context of the rapidly developing, yet still nascent, domain of *AI-augmented solution* development.

Figures 8(a), (b), and (c) in the appendix show the production functions and reveal that countries would input more combinations of *tech-labor* and *AI-capital* resources to increase the production of *AI-augmented solutions* if they use all three production functions. In each figure panel, the isoquant curve shifts toward the upper right quadrant, demonstrating such a case. Notably, technological progress not only pushes the factor price frontier of *AI-augmented solution* production outward, but also should result in a "pronounced tilt" of the shifts toward the resource with a comparative advantage in each economy (Figs. 1 and 2). Unlike classical models, the shifting rays emanating from the origin arise because one of the two resources (either *tech-labor* or *AI-capital*) will be employed in greater quantities ($r\%$), a reflection of its relative affordability or comparative advantage compared to the other over the course of time.

To account for such "pronounced tilt," we can adapt and modify the conventional production functions presented in Eqs. (1), (2) and (3) as follows:

4.1.1 Cobb–Douglas Production Functions (from Eq. (1))

Developed Economy:

$$\overline{q}_{t=t} = t[(1+r)^{(t-1)}K^{\alpha}L^{\beta}] \tag{9}$$

Developing Economy:

$$\overline{q}_{t=t} = t[K^{\alpha}(1+r)^{(t-1)}L^{\beta}] \tag{10}$$

where $\beta = (1-\alpha)$ and $0 \leq \alpha, \beta \leq 1$, and K is *AI-capital*, L is *tech-labor*, and q is the volume of the *AI-augmented solution* created by the combination of inputs.

4.1.2 Linear Production Functions (from Eq. (2))

Developed Economy:

$$\overline{q}_{t=t} = t(\frac{(1+r)^{(t-1)}K}{\alpha} + \frac{L}{\beta}) \tag{11}$$

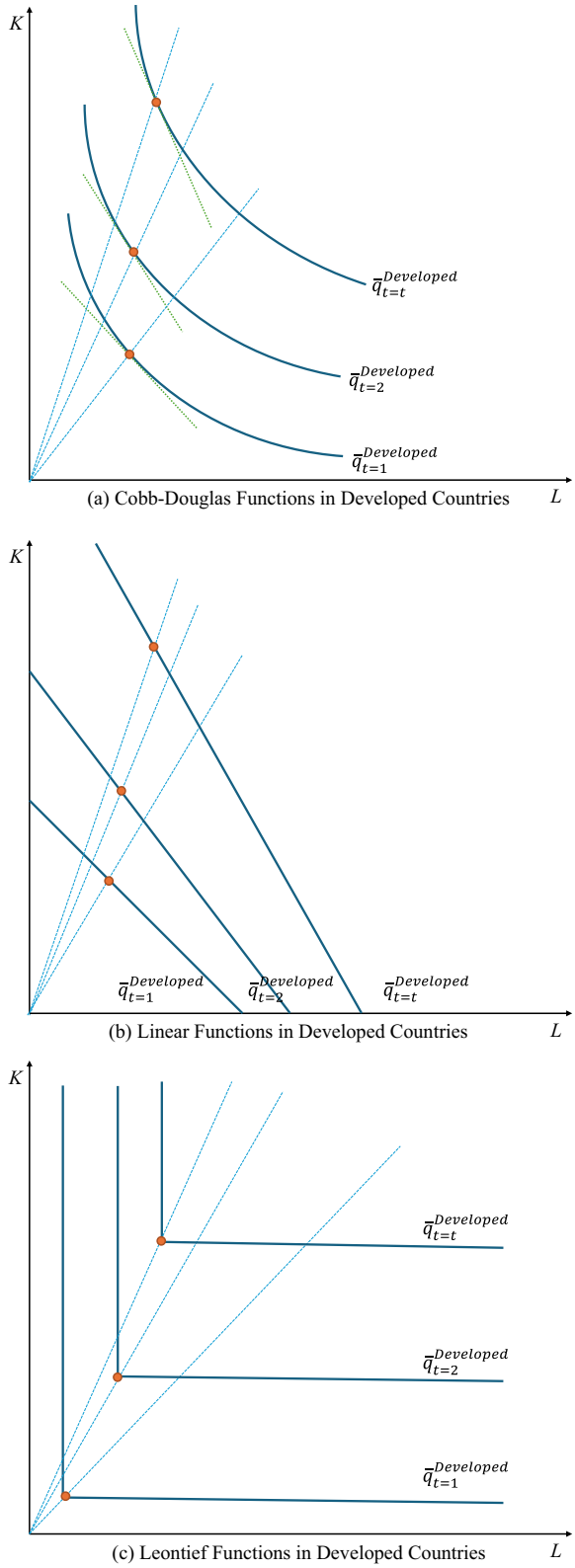


Fig. 1 Production Functions in Developed Countries

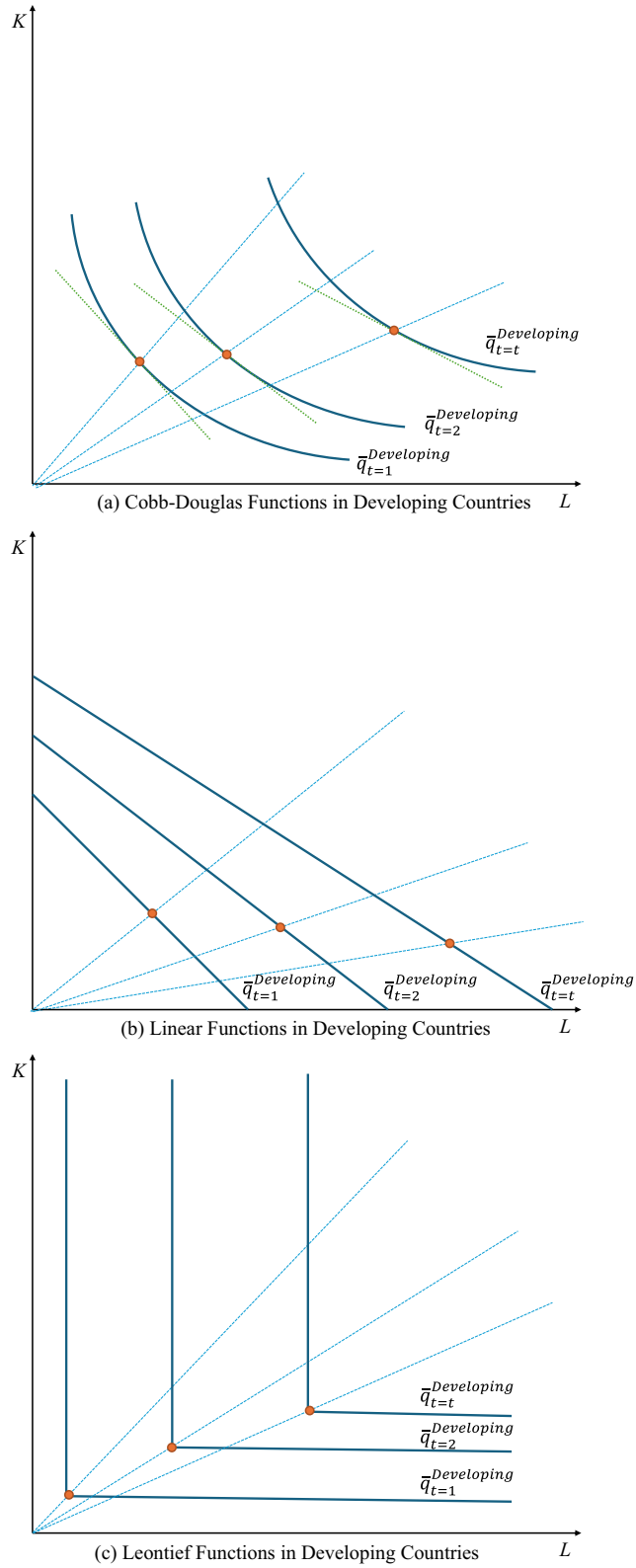


Fig. 2 Production Functions in Developing Countries

Developing Economy:

$$\overline{q}_{t=t} = t(\frac{K}{\alpha} + \frac{(1+r)^{(t-1)}L}{\beta}) \tag{12}$$

where $\beta = (1-\alpha)$ and $0 \leq \alpha, \beta \leq 1$, and K is *AI-capital*, L is *tech-labor*, and q is the volume of the *AI-augmented solution* created by the combination of inputs.

4.1.3 Leontief Production Functions (from Eq. (3))

Developed Economy:

$$\overline{q}_{t=t} = t[\min\left(\frac{K}{\alpha(1+r)^{(t-1)}}, \frac{L}{\beta}\right)] \tag{13}$$

Developing Economy:

$$\overline{q}_{t=t} = t[\min\left(\frac{K}{\alpha}, \frac{L}{\beta(1+r)^{(t-1)}}\right)] \tag{14}$$

where $\beta = (1-\alpha)$ and $0 \leq \alpha, \beta \leq 1$, and K is *AI-capital*, L is *tech-labor*, and q is the volume of the *AI-augmented solution* created by the combination of inputs.

Figures 1(a), (b), and (c) delineate phenomena whereby each of the production functions of the developed country moves outward. In addition, each of the production functions moves on a different ray extending from the origin over time (the rays move toward the upper left side over time). Therefore, each of the production functions (Cobb–Douglas and Linear functions) exhibits an increasingly steep upward trajectory over time. This observation underscores the presence of a competitive advantage in *AI-capital* among developed countries, facilitating their capacity to use *AI-capital* as a strategic asset. This, in turn, significantly enhances their productive capabilities in a progressive manner.

Figures 2(a), 2(b), and 2(c) illustrate scenarios wherein the production functions of developing countries also exhibit an outward shift. In contrast to the patterns observed within developed countries, however, each production function traverses a distinct ray emanating from the origin, with the ray shifting toward the lower right side over time. Therefore, each production functions (Cobb–Douglas and Linear functions) exhibits a decreasing, less-steep downward slope over time. This observation highlights the competitive advantage in *tech-labor* possessed by developing countries, enabling them to employ *tech-labor* as a strategic resource. Consequently, this advantage facilitates a substantial and progressive enhancement of their production capabilities.

Note that the traditional Leontief production function (Eq. (3)) is represented as $\overline{q}_{t=1} = t[\min\left(\frac{K}{\alpha}, \frac{L}{\beta}\right)]$. This implies that the input quantities of *AI-capital* (K) and *tech-labor* (L) must maintain a fixed ratio of $\alpha : \beta$ to produce one (1) unit of *AI-augmented solution*.[5] In this study, we assume that each country possesses a distinct comparative advantage in resources. Therefore, each country increasingly uses its comparative advantage resource to produce the *AI-augmented solution* over time. Unlike the previous two production functions, Cobb–Douglas and Linear, the Leontief production function in our analysis adjusts the values of α and β over time. For instance, in the case of the developed economy, α at 1st iteration → $(1+r)\alpha$ at 2nd iteration → $(1+r)^{(t-1)}\alpha$ at nth iteration, while for the developing economy, β at 1st iteration → $(1+r)\beta$ at 2nd iteration → $(1+r)^{(t-1)}\beta$ at nth iteration, as depicted in Eqs. (13) and (14).

4.2 Simulation for Developed and Developing Economies

A salient concept encapsulated within Eqs. (9) to (14) revolves around the notion of "surge in demand," symbolized as $(1+r)^{(t-1)}$. This concept arises from our examination of the gradual shift in the value of one of the production factors, either *tech-labor* or *AI-capital* resources, over time. Consequently, as countries endeavor to amplify their outputs of the *AI-augmented solution*, they tend to employ a larger quantity (r% more) of the resource in which they possess a comparative advantage. This phenomenon is designated as the "surge in demand."

For instance, consider a scenario in which a developed country initially allocates 1,000 units of *tech-labor* and 4,000 units of *AI-capital*, employing the Cobb–Douglas function to produce 2,000 units of the *AI-augmented solution* ($\alpha = \beta = 0.5$). Given the comparative advantage of *AI-capital* in comparison to *tech-labor* resources, and the ongoing rise in the prominence of AI, this economy has the potential to expand its *AI-capital* resources, perhaps by 10% ($r = 0.1$), to increase the quantity of the *AI-augmented solution*.

Consequently, the developed economy can now mobilize 2,000 units of *tech-labor* (comprising the initial 1,000 units, multiplied by a temporal iteration factor of 2) and 8,800 units of *AI-capital* (originating from the initial 4,000 units, multiplied by a temporal iteration factor of 2, and adjusted for a 10% surge in demand). Such an adjustment leads to the production of 4,195 units of the *AI-augmented solution*, contrasting with the initial allocation of 8,000 units of *AI-capital*, which would produce 4,000 units of the *AI-augmented solution* during the second iteration. This progression implies that the required quantities of *tech-labor* and *AI-capital* at the time of the tth iteration are $t \times L$ and $t \times (1+r)^{(t-1)} \times K$, respectively (see Table 1(a)).

It is important to note that our intention here is not to determine the validity of assuming a 10% surge in *AI-capital*

[5] $\overline{q}_{t=1} = t[\min\left(\frac{K}{\alpha}, \frac{L}{\beta}\right)] \rightarrow (K^*_{t=1} = \alpha \text{ and } L^*_{t=1} = \beta) \rightarrow \overline{q}_{t=1} = [\min\left(\frac{\alpha}{\alpha}, \frac{\beta}{\beta}\right)] = \min(1, 1) = 1$

demand, nor is it of significance what the initial levels of *tech-labor* and *AI-capital* endowments are. Instead, our aim in presenting the simulation results in Table 1(a) is to underscore the critical importance of accounting for surge demand within the context of the subsequent analyses presented in this paper.

In Table 1(a), another thing which we want to highlight the slope. In Eq. (6), we prove $MRTS_{LK}$ is the slope of each isoquant. It is denoted as $-\frac{\partial f/\partial L}{\partial f/\partial K}$. We can then solve Eq. (9) to check the slope of the isoquant given the combinations of resources in the simulation of Table 1(a).

Starting from Eqs. (6) in Section 3 and (9) in Section 4.1, we obtain,

$$\partial f/\partial L = t[(1+r)^{(t-1)}K^{\alpha}\beta L^{(\beta-1)}] \tag{15}$$

$$\partial f/\partial K = t[(1+r)^{(t-1)}\alpha K^{(\alpha-1)}L^{\beta}] \tag{16}$$

$$\therefore \frac{dK}{dL} = -\frac{\partial f/\partial L}{\partial f/\partial K} = -\frac{\beta}{\alpha}\frac{K}{L} \tag{17}$$

Consequently, as per our assumption, the slope of the isoquant for the developed economy, delineating combinations of resources, exhibits a progressive increase over time (see Table 1(a)). This phenomenon should be discernible in Fig. 1(a).

Similarly, let us consider a scenario wherein a developing country initially allocates 5,000 units of *tech-labor* and 2,000 units of *AI-capital*, employing the Cobb–Douglas production function to produce 3,162 units of the *AI-augmented solution*. Given the cost-effectiveness of *tech-labor* in comparison to *AI-capital* resources and the increasing prominence of AI, this economy has the potential to augment its *tech-labor* force by, for instance, 4% ($r=0.04$) to increase *AI-augmented solution* production.

Consequently, the developing economy can now engage 10,400 units of *tech-labor* (comprising the initial 5,000 units, multiplied by a temporal iteration factor of 2, and adjusted for a 4% surge in demand), as opposed to the initial 10,000 units of *tech-labor*, alongside 4,000 units of *AI-capital* (derived from the initial 2,000 units, multiplied by a temporal iteration factor of 2). This modification results in the generation of 6,450 units of the *AI-augmented solution*. The simulation implies that the requisite units of *tech-labor* and *AI-capital* at the tth iteration are determined by $t \times (1.04)^{(t-1)} \times L$ and $t \times K$, respectively. This relationship is visually represented in Table 1(b).

Again, the intention here is not to assess the validity of a 4% surge in *tech-labor* demand. Instead, the presentation of the simulation results in Table 1(b) serves to underscore the importance of incorporating surge in demand into the subsequent analyses presented in this paper. Furthermore,

Table 1 Simulation of Cobb–Douglas Production Function

Panel A For Developed Country

Time: (t)	Surge in Demand: $(1+r)^{(t-1)}$	Quantity: $\bar{q}_{t=t}$	*AI-capital* Inputs: $K^*_{t=t}$	α:	*tech-labor* Inputs: $L^*_{t=t}$	β:$(1-\alpha)$	Slope:
1	1.000	2.000	4.000	0.500	1.000	0.500	–4.000
2	1.100	4.195	8.800	0.500	2.000	0.500	–4.400
3	1.210	6.600	14.520	0.500	3.000	0.500	–4.840
4	1.331	9.230	21.296	0.500	4.000	0.500	–5.324
5	1.464	12.100	29.282	0.500	5.000	0.500	–5.856

Notes:r= 10%, initial *AI-capital* (K) input = 4, initial *tech-labor* (L) input = 1,α=β=0.5, and t = {1, 2, 3, 4, 5}
Also, K is in \$1,000, L is in 1,000 people, and q is in 1,000 solutions

Panel B For Developing Country

Time: (t)	Surge in Demand: $(1+r)^{(t-1)}$	Quantity: $\bar{q}_{t=t}$	*AI-capital* Inputs: $K^*_{t=t}$	α:	*tech-labor* Inputs: $L^*_{t=t}$	β:$(1-\alpha)$	Slope:
1	1.000	3.162	2.000	0.500	5.000	0.500	–0.400
2	1.100	6.450	4.000	0.500	10.400	0.500	–0.385
3	1.210	9.866	6.000	0.500	16.224	0.500	–0.370
4	1.331	13.416	8.000	0.500	22.497	0.500	–0.356
5	1.464	17.102	10.000	0.500	29.246	0.500	–0.342

Notes:r= 4%, initial *AI-capital* (K) input = 2, initial *tech-labor* (L) input = 5,α=β=0.5, and t = {1, 2, 3, 4, 5}
Also, K is in \$1,000, L is in 1,000 people, and q is in 1,000 solutions

Note that our intention here is not to determine the validity of assuming a 10% surge in *AI-capital* demand, nor is it of significance what the initial levels of *tech-labor* and *AI-capital* endowments are. Instead, our aim in presenting the simulation results in Table 1(a) is to underscore the critical importance of accounting for surge in demand within the context of the subsequent analyses presented in this paper.

the assumption of the surge in demand forms a fundamental prerequisite for our analytical framework.

To elucidate the gradual reduction in the isoquant slope within the developing country, as depicted in Fig. 2(a), we can derive Eqs. (6) and (10) as follows:

Starting from Eqs. (6) in Section 3 and (10) in Section 4.1, we obtain,

$$\partial f/\partial L = t[K^{\alpha}\beta(1+r)^{(t-1)}L^{(\beta-1)}] \tag{18}$$

$$\partial f/\partial K = t[\alpha K^{(\alpha-1)}(1+r)^{(t-1)}L^{\beta}] \tag{19}$$

$$\therefore \frac{dK}{dL} = -\frac{\partial f/\partial L}{\partial f/\partial K} = -\frac{\beta}{\alpha}\frac{K}{L} \tag{20}$$

Furthermore, we demonstrate that the slope of the production function for the developing economy, depicting combinations of resources, diminishes over time, as illustrated in Table 1(b). This phenomenon is observable in Fig. 2(a).

Countries identified as either developed or developing, when employing Linear production functions in simulations, demonstrate similar patterns of endowment usages for the production of a greater volume of *AI-augmented solution*, akin to those observed with Cobb–Douglas production functions. For instance, let's revisit the scenario presented in a previous simulation, wherein a developed country initially allocates 1,000 units of *tech-labor* and 4,000 units of *AI-capital*.

The developed country uses a Linear production function to produce the *AI-augmented solution*. Therefore, the economy initially produces 10,000 units of the *AI-augmented solution* using the Linear function. Due to the relative cost-effectiveness of *AI-capital* compared to *tech-labor* resources, coupled with the increasing demand for the *AI-augmented solution*, this economy has the capacity to augment its *AI-capital* by, say 10% ($r=0.1$) to boost the volume of the *AI-augmented solution*. Consequently, the developed economy can now employ 2,000 units of *tech-labor* (1,000 units × 2, accounting for temporal iteration) and 8,800 units of *AI-capital* (4,000 units × 2, factoring in the temporal iteration × 1.1, considering surge demand). Such simulation results in the production of 21,600 units of the *AI-augmented solution*. Therefore, similar to the Cobb–Douglas production case, the progression implies that the required quantities of *tech-labor* and *AI-capital* at the time of the *t*th iteration are $t \times L$ and $t \times (1+r)^{(t-1)} \times K$, respectively. And the Linear production function produces $t[L + (1+r)^{(t-1)} \times K]$ of the *AI-augmented solution* (see Table 2(a)).

As outlined in Table 1(a), Table 2(a) exhibits a similar escalating gradient for the developed country, a phenomenon evident in Fig. 1(b). The slope of the Linear function can be determined by the following equation:

Starting from Eq. (11) in Section 4.1, we obtain,

$$\therefore \frac{dK}{dL} = -[\frac{\beta\bar{q}_{t=t}}{tL} - \frac{\beta}{\alpha}\frac{(1+r\%)^{(t-1)}K}{L}] \tag{21}$$

Table 2 Simulation of Linear Production Function

Panel A For Developed Country

Time: (t)	Surge in Demand: $(1+r)^{(t-1)}$	Quantity: $\bar{q}_{t=t}$	*AI-capital* Inputs: $K^*_{t=t}$	α:	*tech-labor* Inputs: $L^*_{t=t}$	β:$(1-\alpha)$	Slope:
1	1.000	10.000	4.000	0.500	1.000	0.500	–6.000
2	1.100	21.600	8.800	0.500	2.000	0.500	–6.400
3	1.210	35.040	14.520	0.500	3.000	0.500	–6.840
4	1.331	50.592	21.296	0.500	4.000	0.500	–7.324
5	1.464	68.564	29.282	0.500	5.000	0.500	–7.856

Notes:r= 10%, initial *AI-capital* (K) input = 4, initial *tech-labor* (L) input = 1,α=β = 0.5, and t = {1, 2, 3, 4, 5}
Also, K is in \$1,000, L is in 1,000 people, and q is in 1,000 solutions

Panel B For Developing Country

Time: (t)	Surge in Demand: $(1+r)^{(t-1)}$	Quantity: $\bar{q}_{t=t}$	*AI-capital* Inputs: $K^*_{t=t}$	α:	*tech-labor* Inputs: $L^*_{t=t}$	β:$(1-\alpha)$	Slope:
1	1.000	14.000	2.000	0.500	5.000	0.500	–2.400
2	1.100	28.800	4.000	0.500	10.400	0.500	–2.385
3	1.210	44.448	6.000	0.500	16.224	0.500	–2.370
4	1.331	60.995	8.000	0.500	22.497	0.500	–2.356
5	1.464	78.493	10.000	0.500	29.246	0.500	–2.342

Notes:r= 4%, initial *AI-capital* (K) input = 2, initial *tech-labor* (L) input = 5,α=β = 0.5, and t = {1, 2, 3, 4, 5}
Also, K is in \$1,000, L is in 1,000 people, and q is in 1,000 solutions

By applying the same resource endowment and surge in demand scenario for a developing country as used in the previous Cobb–Douglas production function simulation, we can generate $t[K + (1 + r)^{(t-1)} \times L]$ at the time of the tth iteration. Table 2(b) illustrates a decreasing slope for the developing country as time progresses, a trend supported by the evidence in Fig. 2(b). The slope of the Linear function can be calculated using the following equation:

Starting from Eq. (12) in Section 4.1, we obtain,

$$\therefore \frac{dK}{dL} = -[\frac{\beta \overline{q}_{t=t}}{t(1+r)^{(t-1)}L} - \frac{\beta}{\alpha}\frac{K}{(1+r)^{(t-1)}L}] \tag{22}$$

Unlike the Cobb–Douglas and Linear production functions, the elasticity of substitution is zero in Leontief production functions. Therefore, the surge in demand in a Leontief production function extends beyond quantity and maintains a fixed ratio of the two production factors for production. Therefore, the previous simulation scenarios with Cobb–Douglas and Linear production functions need to be modified.

In the Leontief production case, the fix ratio is dependent on the denominators. This means α and β with $r\%$ become discount factors for the endowments. For example, consider a situation of a data center with ML and human operators processing data in a certain format together, and the data center sells a software that contains the processed data as an *AI-augmented solution*.

Initially, the operation may need 2,000 ML units, working in tandem with 2,000 human operators to manage data processing. Over time, however, with advancements in ML capabilities surpassing the output efficiency of human labor by r (e.g., 10% in this simulation), the data center in developed countries could adjust its operational model to rely on 4,000 ML units while hiring less workforce to sustain data processing requirements due to their Leontief production functions. This progression implies that the required quantities of *tech-labor* and *AI-capital* at the time of the tth iteration are $t \times L/\beta$ and $t \times k/\alpha(1+r)^{(t-1)}$, respectively; the simulation results are visually depicted in Table 3(a).

In the described scenario, developing countries might pursue a distinct strategy in resource allocation that prioritizes *tech-labor*. With *tech-labor* costs remaining relatively lower in developing countries, data centers, for instance, might significantly increase the employment of human labor over ML units over time. The outcomes of this strategy, particularly under a scenario applying a 4% discount ratio ($r = 0.04$) as a sample, are detailed in Table 3(b), illustrating the impact of different combinations of endowment inputs of $t \times L/\beta(1+r)^{(t-1)}$ and $t \times K/\alpha$ at the time of the tth iteration in developing countries on the *AI-augmented solution* over time.

4.3 Edgeworth Box of Developed and Developing Economies

To comprehensively examine the evolving dynamics of both developed and developing economies over time, the most effective approach is to employ the Edgeworth box, as detailed in Section 3.3. In Figs. 3(a), (b), and (c), both economies extend their respective isoquants from their original positions toward the upper right corner as they produce larger quantities of the *AI-augmented solution*.

Our analysis indicates that during the initial stages, as depicted in Figs. 3(a), (b), and (c), these two economies abstain from resource trading, despite possessing distinctive advantages in their respective resource endowments. This decision arises from the fact that both economies possess resource surpluses sufficient to accommodate an increase in the production of the *AI-augmented solution*. However, as time progresses, specifically at the tth iteration, the two economies converge at point A in Figs. 3(a) and (b), and at both point A for developing countries and point A" for developed countries in Fig. 3(c), at which juncture resource surpluses are depleted and the total quantities of each resource remain constant over time.

As both economies exhibit comparative advantages in distinct resources (leading to different slopes at point A in Figs. 4(a), (b), and (c)), there exists potential for these economies to enhance their output production by engaging in resource exchange between the two countries. However, regardless of which economy takes the initiative on this resource exchange, such trade or partnership between the two economies comes to fruition only under conditions in which the proposed transaction, represented as adjustments in factor allocations, results in advantages for at least one party without detriment to the other. Therefore, the exclusive zone where resource exchange can take place is confined to the shaded area in Figs. 4(a), (b), and (c).

The analysis of two countries employing Cobb–Douglas production functions, as depicted in Fig. 4(a), unfolds across three scenarios based on who initiates resource trade: (1) mutual initiation by both economies, (2) initiation by the developed economy, and (3) initiation by the developing economy. These scenarios are further explored through visual illustrations in Figs. 5(a), (b), and (c).

Similarly, for two economies applying Linear production functions, represented in Fig. 4(b), the analysis is segmented into three distinct scenarios predicated on the initiating party of the resource trade: (1) both economies initiating, (2) initiation led by the developed economy, and (3) initiation led by the developing economy. The visualization of these scenarios is provided in Figs. 6(a), (b), and (c).

Lastly, the exploration of two nations adopting Leontief production functions, showcased in Fig. 4(c), is conducted

Table 3 Simulation of Leontief Production Function

Panel A For Developed Country

Time: (t)	Discount Rate: $(1+r)^{(t-1)}$	Quantity: $\bar{q}_{t=t}$	*AI-capital* Inputs: $K^*_{t=t}$	α:	*tech-labor* Inputs: $L^*_{t=t}$	β:$(1-\alpha)$	Slope:
1	1.000	4.000	2.000	0.500	2.000	0.500	N/A
2	1.100	14.545	4.000	0.500	3.636	0.500	N/A
3	1.210	29.752	6.000	0.500	4.958	0.500	N/A
4	1.331	48.084	8.000	0.500	6.010	0.500	N/A
5	1.464	68.301	10.000	0.500	6.830	0.500	N/A

Notes:r= 10%, initial *AI-capital* (K) input = 2, initial *tech-labor* (L) input = 2,α=β = 0.5, and t = {1, 2, 3, 4, 5}
Also, K is in $1,000, L is in 1,000 people, and q is in 1,000 solutions

Panel B For Developing Country

Time: (t)	Discount Rate: $(1+r)^{(t-1)}$	Quantity: $\bar{q}_{t=t}$	*AI-capital* Inputs: $K^*_{t=t}$	α:	*tech-labor* Inputs: $L^*_{t=t}$	β:$(1-\alpha)$	Slope:
1	1.000	4.000	2.000	0.500	2.000	0.500	N/A
2	1.100	15.385	3.846	0.500	4.000	0.500	N/A
3	1.210	33.284	5.547	0.500	6.000	0.500	N/A
4	1.331	56.895	7.112	0.500	8.000	0.500	N/A
5	1.464	85.480	8.548	0.500	10.000	0.500	N/A

Notes:r= 4%, initial *AI-capital* (K) input = 2, initial *tech-labor* (L) input = 2, initialα=β = 0.5, and t = {1, 2, 3, 4, 5}
Also, K is in $1,000, L is in 1,000 people, and q is in 1,000 solutions

within three specific scenarios based on the initiator of the resource trade: (1) a joint effort by both economies, (2) initiated by the developed economy, and (3) initiated by the developing economy. The graphical representation of these scenarios is displayed in Figs. 7(a), (b) and (c).

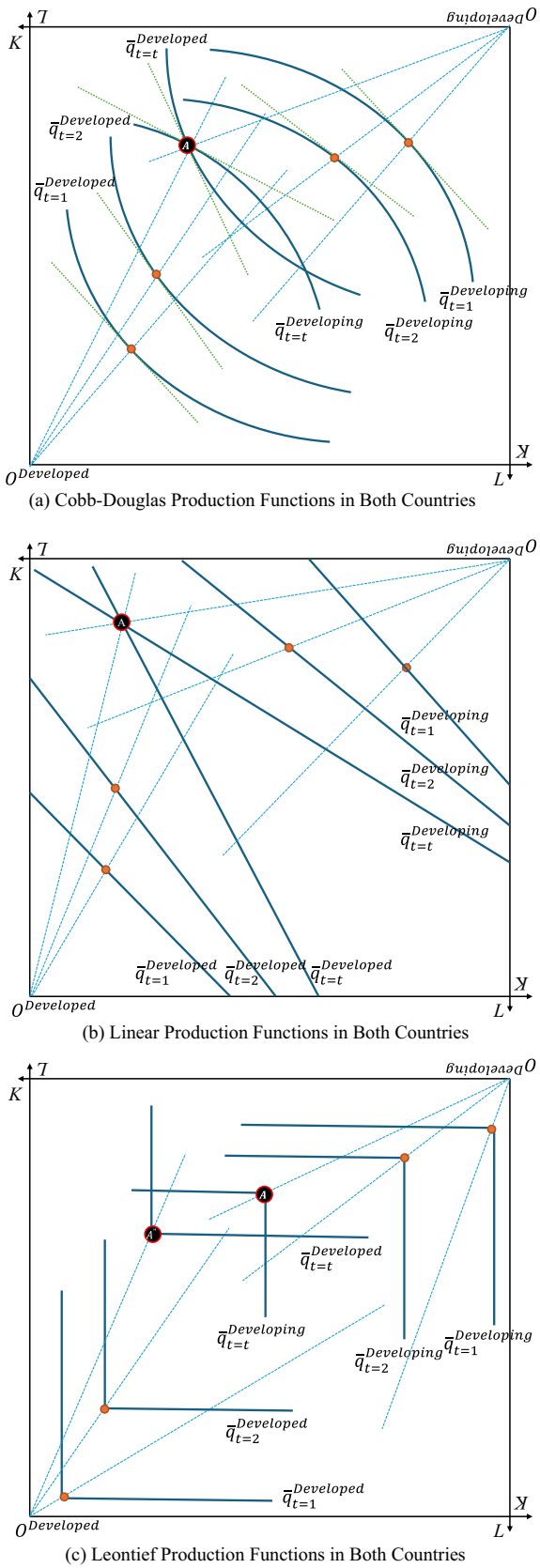
4.4 Resource Exchanges of Developed and Developing Economies

4.4.1 Pareto Improvement and Pareto Optimality using Cobb–Douglas Functions

In Fig. 5(a), both economies initiate the trade together and shift their isoquants away from their respective starting positions (origins). Consequently, any arbitrary point denoted as B within the initial shaded area outlined in Fig. 4(a) would constitute a Pareto improvement for both countries. Intriguingly, the two isoquants converge at point B in Fig. 5(a), rendering it a point of tangency with an equivalent slope ($MRTS_{LK}$).

Figure 5(b) illustrates a scenario in which the developed country initiates resource trade. Given the developed country's comparative disadvantage in *tech-labor*, *tech-labor* moves to the developed country and capital flows to the developing country. Graphically, this situation is depicted as the isoquant for the developed country shifting toward the upper right side, while the isoquant for the developing country remains stationary. Similar to Fig. 5(a), the isoquant for the developed country ceases its movement when it becomes tangent to the isoquant for the developing country at point C. At this juncture, the slopes of both isoquants align. Furthermore, the original isoquant for the developed country ($\bar{q}^{Developed}_{t=t}$) is closer to its origin than the shifted isoquant ($\bar{q}^{Developed''}_{t=t}$). Conversely, the isoquant of the developing economy remains in its original position ($\bar{q}^{Developing}_{t=t}$). Therefore, point C serves as a point of Pareto improvement. Simultaneously, point C attains Pareto optimality since any alternative points, aside from C, would not provide opportunities for Pareto improvement through resource allocation.

In the final scenario depicted in Fig. 5(c), it is the developing country that instigates the resource trade. Visually, this scenario is shown by the shifting of the isoquant for the developing country toward the lower left side (which is the upper right relative to the developing country's origin), while the isoquant for the developed country remains static. Analogous to Figs. 5(a) and (b), the movement of the developing country's isoquant ceases when it becomes tangent to the isoquant for the developed country at point D. At this juncture, the slopes of both isoquants coincide. Furthermore, the original isoquant for the developing country ($\bar{q}^{Developing}_{t=t}$) is closer to its origin compared to the shifted isoquant ($\bar{q}^{Developing''}_{t=t}$) as seen in Fig. 5(c). Conversely, the isoquant for the developed economy remains in its original position ($\bar{q}^{Developed}_{t=t}$). Consequently, point D emerges as a locus of Pareto improvement. Simultaneously, point D attains Pareto

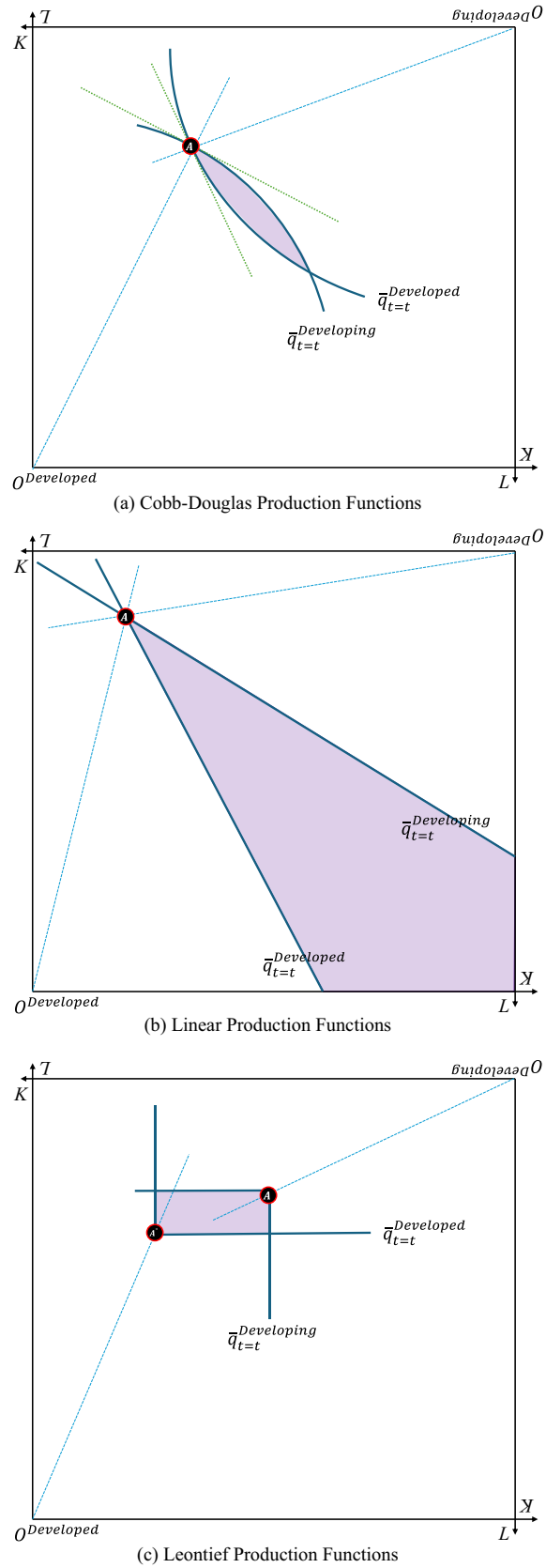


(a) Cobb-Douglas Production Functions in Both Countries

(b) Linear Production Functions in Both Countries

(c) Leontief Production Functions in Both Countries

Fig. 3 Temporal Evolution of Production Functions in Both Countries



(a) Cobb-Douglas Production Functions

(b) Linear Production Functions

(c) Leontief Production Functions

Fig. 4 Potential Areas for Attaining Pareto Improvement

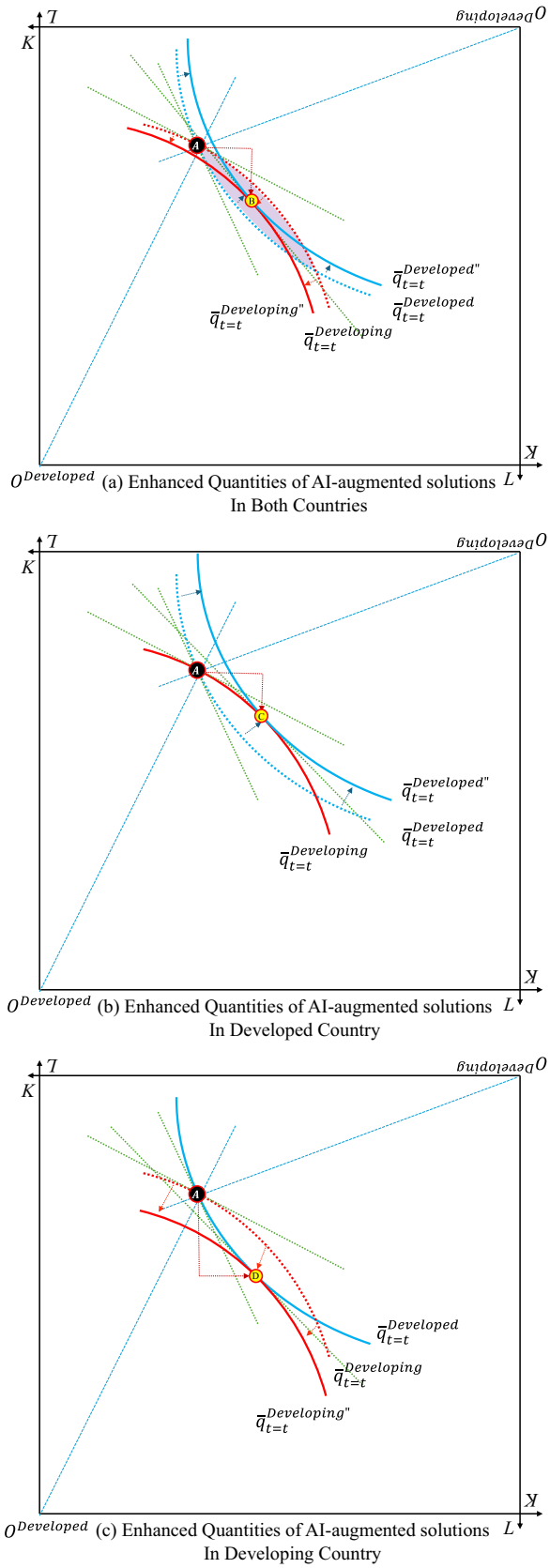


Fig. 5 Output Improvement, Pareto Improvement, and Pareto Optimality (C-D Function)

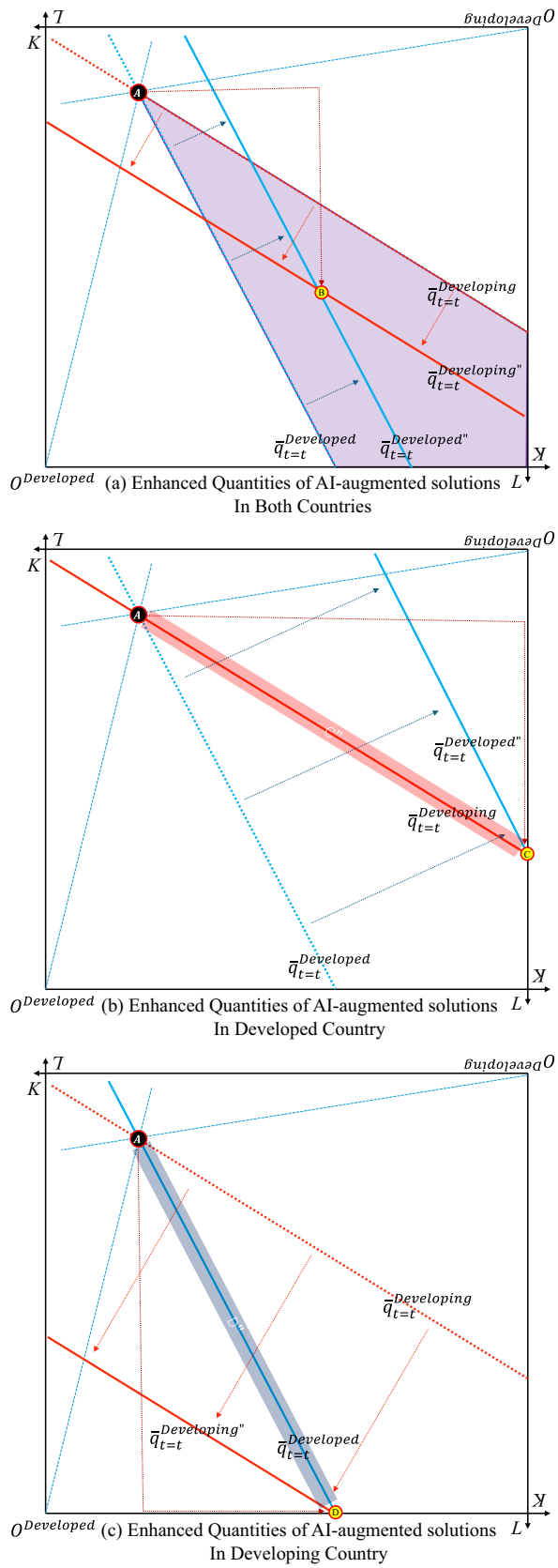


Fig. 6 Output Improvement, Pareto Improvement, and Pareto Optimality (Linear Function)

optimality, as any alternative points, apart from D, would not provide opportunities for Pareto improvement through resource allocation.

4.4.2 Pareto Improvement and Pareto Optimality using Linear Production Functions

In Section 4.4.1, we presented the rationale and justifications for resource trade. These rationales and justifications remain consistent in this section. Consequently, we will abbreviate the explanations in this section.

In Fig. 6(a), both economies engage in simultaneous trade, prompting the shift of their isoquants away from their initial positions (origins). As a result, any point, symbolized as B, located in the initial shaded area in Fig. 4(b) represents a Pareto improvement for both countries. At point B, the two isoquants intersect.

In Fig. 6(b), the developed country takes the lead in resource trade. Visually, this scenario is represented by the isoquant for the developed country shifting toward the upper right, while the isoquant for the developing country remains stable. As in Fig. 6(a), the isoquant for the developed country halts its movement upon intersecting with the isoquant for the developing country at point C. Notably, point C also signifies the *AI-capital* intercept for the developing economy.

In Fig. 6(c), the developing country takes the initiative in resource trade. Visually, this scenario is represented by the isoquant for the developing country shifting toward the lower left (which is the upper right relative to the developing country's origin), while the isoquant for the developed country remains stable. Similar to Figs. 6(a) and (b), the movement of the developing country's isoquant halts when it intersects with the isoquant for the developed country at point D. Notably, point D also represents the *tech-labor* intercept for the developed economy.

In the scenarios in Figs. 6(b) and (c), both economies converge with the other's isoquants at positions denoted as the *AI-capital* intercept for the developing country and the *tech-labor* intercept for the developed country. Each of these isoquants comes to a standstill because these points, labeled C and D, represent the only locations characterized by Pareto optimality. For instance, in the scenario depicted in Fig. 6(b), the developed country might cease its movement at the point denoted as C, which falls between points A and C. Subsequently, other points (such as C ″) can be identified that can increase *AI-augmented solution* outputs through resource exchange without any detriment to the outputs of the developing economy. Similarly, in the scenario illustrated in Fig. 6(c), the developing country may halt at the point labeled as D, situated between points A and D. Consequently, additional points (like D") can be identified that can increase *AI-augmented solution* outputs through

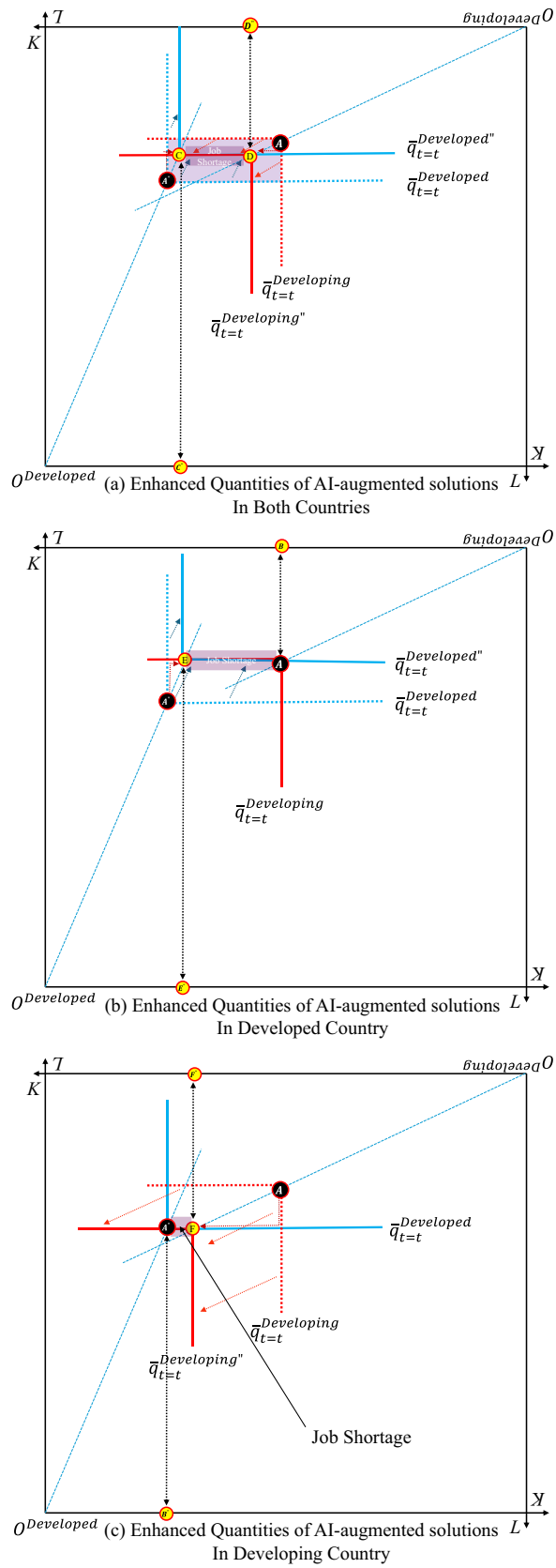


Fig. 7 Output Improvement, Pareto Improvement, and Pareto Optimality (Leontief Function)

resource exchange without adversely affecting the outputs of the developed economy.

This phenomenon can be attributed to the specific characteristics of a Linear production function. As indicated by Eq. (8) in Section 3.2, we denote the substitution elasticity (σ) of a Linear production function as infinite (∞), signifying that a Linear production function aligns with the concept of a *perfect factor substitution production function*.

As a result, the developed economy is capable of employing all available *tech-labor* resources from the developing country (see point C in Fig. 6(b)) at a reduced cost, thereby having the potential to substitute these for its *AI-capital* resources. Consequently, while the developing country may experience an influx of capital, it could also face a complete outflow of its *tech-labor* resources. Conversely, by leveraging its comparative advantage in *tech-labor*, the developing country has the potential to attract all available *AI-capital* resources from the developed economy (see point D in Fig. 6(c)).

4.4.3 Pareto Improvement and Pareto Optimality using Leontief Production Functions

In Fig. 7(a), both economies engage in simultaneous trade, leading to a shift of their isoquants from their initial positions (origins). Consequently, any points denoted as C for the developed country and D for the developing country, located in the initial shaded area in Fig. 4(c), denote a Pareto improvement for both countries.

An important observation to note is that prior to the convergence of the two countries at points A and A", they undergo shifts in their Leontief production functions achieved by increasing resource inputs. Nonetheless, once convergence is achieved at points C and D, the opportunity for both economies to enhance the production volume of the *AI-augmented solution* without negatively impacting each other ceases to exist. Moreover, the gap between C and D (see Fig. 7(a)) symbolizes potential job shortages, indicating that while developed countries may seek to expand their workforce to complement their burgeoning *AI-capital*, such recruitment of *tech-labor* from the developed countries could detrimentally affect the developing countries. On the flip side, this scenario also suggests that while developing countries have the potential to increase their *tech-labor* force, the scarcity of *AI-capital* means that many tech workers may remain unemployed, highlighting a misalignment between labor availability and capital resources.

Figure 7(b) depicts a scenario in which the developed country initiates resource trade. Visually, this scenario manifests as a shift of the Leontief isoquant for the developed country toward the upper right, while the isoquant for the developing country remains stable.

Consequently, the developed country adjusts its optimal resource allocation from point A″ to point E, achieving Pareto optimality. Similar to the scenario in Fig. 7(a), the gap between E and A signifies *tech-labor* unemployment in the developed economy, and the wider gap compared to Fig. 7(a) signifies greater unemployment. We will present a comprehensive account of these findings and engage in further discussion in the next sections.

In the last scenario, depicted in Fig. 7(c), it is the developing country that initiates resource trade. Visually, this scenario manifests as a shift of the isoquant for the developing country toward the lower left (which is the upper right relative to the developing country's origin, $Origin^{Developing}$), while the isoquant for the developed country remains stable. Consequently, the developing country adjusts its optimal resource allocation from point A to point F, achieving Pareto optimality. In line with the scenarios in Figs. 7(a) and (b), the smaller gap between F and A″ signifies *tech-labor* unemployment within the developing country.

To facilitate analytical efficiency and the interpretation of results, we incorporate classical economic constraints, as elaborated in the theory section. In the context of heterogeneous economies and a duality of resources (*tech-labor* and *AI-capital*), we anticipate each economy having distinct reactions to the evolving price dynamics associated with these two factors over time to produce a large volume of a single *AI-augmented solution*. Consequently, we anticipate a relative depreciation of one factor, either within *tech-labor* or *AI-capital* resources, as time progresses. This leads to the expectation that, in their endeavor to increase the volume of the *AI-augmented solution*, each country will tend to use a greater proportion ($0 \leq r \leq 1$) of the resource in which it possesses a comparative advantage.

As time progresses and the prominence of AI technology continues to grow, both economies may show a shared inclination to increase their overall production of *AI-augmented solutions*. Consequently, each country aspires to carefully determine the optimal combination of factor inputs to bolster its production capabilities. In the preliminary phases of our analysis, the two economies abstain from engaging in resource trade, even though they possess unique advantages in their respective resource endowments. This decision is rooted in the fact that both economies maintain surpluses of resources sufficient to accommodate an increase in the production volume of the *AI-augmented solution*. However, as our analysis unfolds, a shift occurs in which the two economies begin to explore opportunities for resource exchange, capitalizing on their comparative advantages to improve the outcomes of the *AI-augmented solution*.

Consequently, in the analytical section, we demonstrate that trade initiation between these two economies materializes only when the proposed trade, involving adjustments in factor allocations, results in advantages for at least one party without causing harm to the other (Pareto improvement). These outcomes are displayed in Table 4.

In Table 4, the first scenario where both economies initiate the trade together causes their isoquants to shift away

Table 4 The Changes in AI-Augmented Solutions

Resource Reallocation	Concurrent Collaboration between the Two Countries		Developed Country Initiates		Developing Country Initiates	
Production Function	Developed Economy	Developing Economy	Developed Economy	Developing Economy	Developed Economy	Developing Economy
Cobb–Douglas	Increase in $q_{t=t}^{*,Developed}$	Increase in $q_{t=t}^{*,Developing}$	Increase in $q_{t=t}^{*,Developed}$	No Change	No Change	Increase in $q_{t=t}^{*,Developing}$
Linear	Increase in $q_{t=t}^{*,Developed}$	Increase in $q_{t=t}^{*,Developing}$	Increase in $q_{t=t}^{*,Developed}$	No Change	No Change	Increase in $q_{t=t}^{*,Developing}$
Leontief	Increase in $q_{t=t}^{*,Developed}$	Increase in $q_{t=t}^{*,Developing}$	Increase in $q_{t=t}^{*,Developed}$	No Change	No Change	Increase in $q_{t=t}^{*,Developing}$

from their initial positions (origins) (see Figs. 5(a), 6(a), and 7(a)). This collaborative effort enables both countries to collectively increase their output of the *AI-augmented solution* ($\overline{q}_{t=t}^{Developed} < \overline{q}_{t=t}^{Developed''}$ and $\overline{q}_{t=t}^{Developing} < \overline{q}_{t=t}^{Developing''}$) by reallocating their resources, regardless of the production functions employed.

In the second scenario, when the developed economy instigates resource trade, the isoquants for the developed country shift toward the upper right quadrant, while the isoquants for the developing country remain stable (see Figs. 5(b), 6(b), and 7(b)). Consequently, this results in an increased volume of the *AI-augmented solution* for the developed economy, while the developing economy maintains the same level of *AI-augmented solution* output ($\overline{q}_{t=t}^{Developed} < \overline{q}_{t=t}^{Developed''}$, but $\overline{q}_{t=t}^{Devoloping}$ stays).

Finally, the third scenario represents the developing economy initiating resource trade. In this scenario, the isoquants for the developing country shift toward the lower left quadrant (or upper right from the developing country's origin ($Origin^{Developing}$)), while the isoquants for the developed country remain stable (see Figs. 5(c), 6(c), and 7(c)). This means that the developing country achieves more volume of *AI-augmented solution* creation than than those produced by the developed economy at an equivalent volume ($\overline{q}_{t=t}^{Developed}$ stays, but $\overline{q}_{t=t}^{Developing} < \overline{q}_{t=t}^{Developing''}$).

Following our examination of the analytical model, our focus shifts toward the potential implications of our findings. The next section is a discussion that examines the broader effects of resource reallocation on each economy, based on the results of our analysis.

5 Discussion of Findings

Building on our analysis of *AI-augmented solution* production changes, we now explore the broader implications of these shifts. Table 5 outlines the potential outcomes of resource reallocation for each economy, setting the stage for an in-depth discussion on the implications of these strategic exchanges.

The first scenario of interest involves concurrent collaboration between both economies. In this context, the developed country can lure *tech-labor* from the developing country with comparatively higher wages, which are still lower than the wages needed to hire *tech-labor* in the developed country. Therefore, the developing economy contends with a "brain drain" phenomenon as tech-workers move away to work in the developed country or take jobs with companies based in the developed country. However, the developing country gains from foreign direct investment (FDI), leading to the generation of new employment opportunities, increased income, and more of the *AI-augmented solution*, all while mitigating risks. In this scenario, in the case of both countries employing Leontief production functions, they encounter a significant shortage of jobs (see Fig. 7(a)) despite economic growth. Additionally, *AI-capital* becomes scarce in both countries (globally, no more residual resources of *AI-capital* exist: The distance between C and C" + the distance between D and D" = $\overline{K}^{Total}$ in Fig. 7(a)), making it challenging to increase the quantity of *AI-capital* units. Consequently, this scarcity may lead to unemployment among skilled *tech-labor* in both countries, as illustrated in Table 5.

The second set of results is related to the context in which the developed country initiates the reallocation of resources. When the developed economy employs the Cobb–Douglas production function, it attracts *tech-labor* from the developing country. This may lead to an increase in work-related visas, such as H1B visas in the United States, resulting in an increased volume of the *AI-augmented solution* for the developed economy. However, in this case, *AI-capital* transfer to the developing country does not take the form of FDI. Instead, it primarily consists of wealth transferred from the developed economy by the returning *tech-labor* force and high-level worker remittances sent from the developed country by those who

Table 5 Possible Outcomes in Each Economy

Resource Reallocation	Concurrent Collaboration between the Two Countries		Developed Country Initiates		Developing Country Initiates	
Production Function	Developed Economy	Developing Economy	Developed Economy	Developing Economy	Developed Economy	Developing Economy
Cobb–Douglas	*Tech-labor* migration from developing countries	FDI	*Tech-labor* migration from developing countries	Worker remittances inflow	*Tech-labor* migration from developing countries	FDI
Linear	*Tech-labor* migration from developing countries	FDI	Attract large scale *tech-labor* from developing economies Huge worker remittances outflow from developed economy	Complete "brain drain" in developing economies	*Tech-labor* migration from developing countries	"Great Transfer of Wealth" to "New World" Huge FDI from developed countries Economic boom in developing countries
Leontief	Shortages of AI-related jobs in both countries	Shortages of AI-related jobs in both countries	Small job creation Unemployment challenges faced by immigrants in developed countries	Huge shortage of AI-related jobs in developing countries	Small shortage of AI-related jobs in developed countries	More jobs, economic boom in developing economy

remain abroad. Such *AI-capital* and its spillovers can be reinvested in various forms, including education, tangible AI assets like high-speed network infrastructure and high-performance computer centers, or intangible digital assets such as data resources. Consequently, *AI-capital* helps bridge the gap caused by a shortage of *tech-labor* in the developing economy, maintaining the same level of *AI-augmented solutions* (see Fig. 5(b)).

Unlike the Cobb–Douglas production functions, Linear production functions are characterized as *perfect factor substitution production functions*, meaning these countries face no challenges in substituting one production factor for another. Therefore, if developed countries equipped with Linear production functions initiate resource trade; the developed economy may allocate idle monetary resources to attract large scale *tech-labor* from developing economies. In this case, the developed economy might issue specialized work-related visas, such as H1B visas for STEM professionals in the United States. The recruiting of *tech-labor* from the developing country would not stop until the developed economy has transferred all residual *AI-capital* into *tech-labor* (at point C of Fig. 6(b)). Consequently, the developing economy may experience a significant "brain drain" issue, but it could also receive substantial worker remittances as compensation. This financial inflow may then be reinvested in *AI-capital* or used to stimulate local markets, establish smart factories, and maintain the same level of the *AI-augmented solution* in the developing economy.

The final situation of this second scenario involves countries with Leontief production functions for *AI-augmented solutions*. In this case, since the developed economy initiates the resource trade, the country can heavily invest in *AI-capital* in the local economy and simultaneously hire only a small number of *tech-labor* workers from the developing country. This is because a Leontief production function demands a fixed resource ratio to generate outcomes, and the developed economy possesses a high input ratio of *AI-capital* to *tech-labor*. Conversely, the Leontief production function of the developing economy features a low input ratio of *AI-capital* to *tech-labor*, resulting in a surplus of *tech-labor* that may lead to unemployment in the developing economy (see Fig. 7(b)). The job shortage in the developing country presents a more serious problem than the situation of Fig. 7(a).

The final series of results illustrated in Table 5 relates to the circumstance wherein the developing economy initiates resource trade. In this third scenario, the developing economy attains more favorable *AI-augmented solution* results compared to those generated by the developed economy at an equivalent volume.

As in the first scenario, when the countries employ the Cobb–Douglas production function, the narrative closely

aligns with the patterns observed in other Cobb–Douglas production function scenarios in Table 5. For instance, developing countries attract FDI from developed nations due to their comparative advantages, such as cost-effective *tech-labor* and a relative scarcity of *AI-capital*. Initiating actions like industrial fairs or foreign-friendly policies, developing nations can draw various forms of FDI, including joint ventures and partnerships with local firms. Such an influx of FDI promotes economic growth, enhances the business environment, and increases *AI-augmented solution* production, while holding the production level of the *AI-augmented solution* constant in the developed countries (see Fig. 5(c)).

In this third scenario, when countries adopt the Linear production function, the developing economy initiating resource trade and leveraging its comparative advantage in *tech-labor* (despite a deficiency in *AI-capital*) becomes strategic. The country may organize industrial fairs or enact policies that encourage foreign businesses and investments to capitalize on this advantage, which occurs primarily because businesses in developed economies find it challenging to hire *tech-labor* due to its high cost. As a result, these businesses may channel significant FDI into the developing country. This influx of FDI can be likened to a "Great Transfer of Wealth" to the "New World," (see Fig. 6(c)), bolstering the local economy of developing countries.

Lastly, when the developing economy uses a Leontief production function and takes the lead in initiating trade of resources, a compelling dynamic unfolds: The developing economy has the capacity to attract FDI, establish joint ventures, provide financial support to local corporations, and engage in mergers and acquisitions by offering various incentives such as tax benefits, hosting industrial fairs, or implementing policies that favor foreign investors. This surge of foreign investments can generate numerous job opportunities and stimulate economic growth in the developing country, resulting in a substantial demand for local labor to occupy these newly created positions (*tech-labor* moves from point A″ to F in Fig. 7(c)). However, on the flip side, there will be limited job creation in the developed economy since a significant portion of its *AI-capital* has been transferred to the developing economy (globally, no more residual resources of *AI-capital* exist, as $AI\text{-}capital = \overline{K}^{Total}$ in Fig. 7(c)). Additionally, as the developing economy, experiencing economic growth, has already filled most of its newly created positions with local *tech-labor*, there may be a minor issue of unemployment in the developed economy (the distance between points A" and F in Fig. 7(c)) due to the scarcity of *AI-capital*.

6 Conclusion, Limitations, and Future Directions

The principal aim of this study is to offer insights into the responses of heterogeneous economies, marked by varying comparative advantages in resource endowments, as they engage in the production of an *AI-augmented solution* over an extended time frame. To explore this research inquiry, we establish a comparative framework involving two markedly dissimilar economies, akin to a developed economy such as the United States and a developing economy such as India. These economies exhibit contrasting levels of *tech-labor* and *AI-capital* resource intensity.

This study has some important implications. First, it provides insights into how two distinct countries can respond to the challenge of increasing the volume of an *AI-augmented solution*. The study offers plausible explanations and explores the potential outcomes (in terms of *AI-augmented solution* volume) under various scenarios, considering two key criteria: (1) the use of three different production functions and (2) the initiation of resource trade under different conditions.

Second, the study sheds light on the potential challenges that each economy might encounter, as elucidated in Section 5. Consequently, it can offer valuable guidance to countries on how to manage their resources effectively. Moreover, policymakers can draw upon these findings to formulate national AI strategies that either safeguard or promote their country's resources to enhance resource use. From a multinational corporation perspective, our study establishes novel viewpoints, laying the foundation for informed decision-making and strategic planning in leveraging *AI-capital* and *tech-labor* resources effectively.

Lastly, this study carves out a fresh trajectory in the research landscape, distinguishing itself from the well-trodden paths typically seen in MIS literature centered around firm-level implications of AI technology. By synthesizing economic perspectives and the discourse on *AI-augmented solutions*, it presents a novel strategic framework for understanding bilateral dynamics between countries at differing stages of economic development. This not only enriches the ongoing discourse but also reveals a stream of potential research avenues in many directions. Leveraging the dynamics of AI as a GPT, it explores a fundamental shift toward collaborative relationships that can be nurtured between developed and developing macroeconomic entities. Thus, it provides a foundational lens for an enriched scholarly assessment that is receptive to divergent economic narratives, thereby promoting a more nuanced and holistic understanding of the AI-induced transformation.

This study has two major inherent limitations. First, it relies on assumptions drawn from economic theory, which can be somewhat idealistic and not always reflective of real-world dynamics. For instance, one of our assumptions posits that the overall quantities of *AI-capital* and *tech-labor* remain constant throughout the temporal scope of our analysis. While such an assumption is common, especially in contexts like the Edgeworth box, the actual number of factors or resources available for production can fluctuate.

Furthermore, this research lacks empirical substantiation for its conclusions. Its primary objective is to offer plausible explanations and potential strategies for heterogeneous economies dealing with resource scarcity while endeavoring to enhance *AI-augmented solutions*. Consequently, the study places a stronger emphasis on providing insights and implications through graphical analyses rather than empirically confirming its hypotheses.

The study offers several potential research directions. First, scholars can use empirical investigation to validate the research's conclusions and explanations. Empirical evidence can bridge the divide between theoretical findings and real-world applications, potentially mitigating some of the limitations stemming from the study's assumptions.

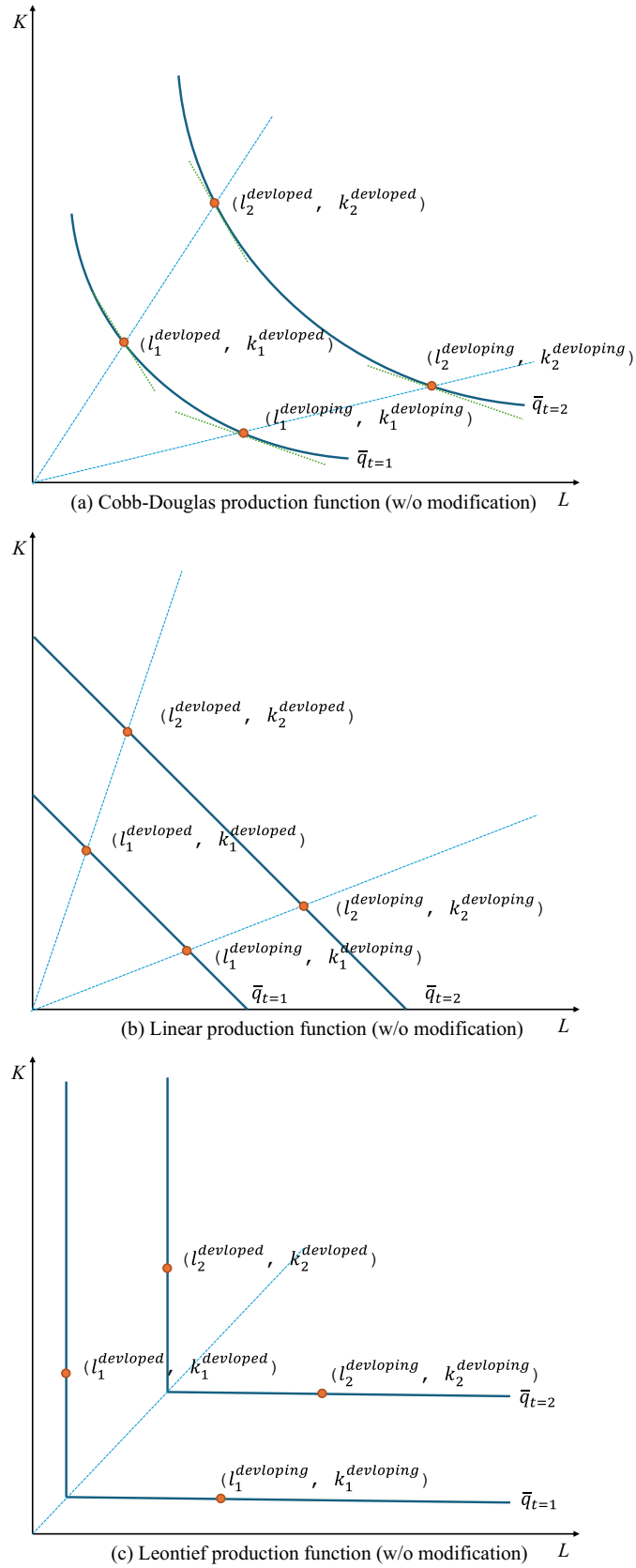
Second, the study relies on a foundational assumption borrowed from classical microeconomics that assumes a constant production function generating identical products across heterogeneous economies. In practice, two distinct countries may employ different production functions even when producing identical goods. Future research could explore the implications of this scenario, delving into the analysis of how two economies using different production functions affect the creation of an *AI-augmented solution*.

Third, the study does not provide mathematical derivations of optimal solution sets when each distinct economy initiates resource trade. Instead, it offers plausible explanations and potential response strategies for heterogeneous economies grappling with resource scarcity while striving to enhance the *AI-augmented solution*. Consequently, the focus is on graphical analysis and interpretation rather than precise solutions derived from equations. Scholars might employ game theory to pinpoint optimal solutions when both countries move simultaneously or when each country moves sequentially. Such research could offer more precise strategies for managing global trade issues in AI production.

Lastly, the study relies on the assumption of a surge in demand. It presumes that each country would utilize more of its comparative advantage resource. However, alternative approaches could be explored. For instance, some economists might consider scenarios where economies with comparative advantages underutilize one of their comparative advantage resources, particularly in the context of automation arguments.

Appendix. Representations of Three Basic Production Functions

Fig. 8 Graphical Representations of Three Production Functions



Acknowledgements We wish to express our profound gratitude for the valuable comments, suggestions, and encouragement bestowed upon us during the ICODO 2023 conference. Moreover, we are appreciative of the invitation to submit our paper for consideration in the special issue of the Information Systems Frontier.

Authors' Contributions
First author: Jin Sik Kim, Ph.D., Assistant Professor
Developed most economic models, analysis, and results of analysis. Wrote first draft of theoretical approach, analysis, and discussion sections. Revised whole paper critically for important intellectual content.
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Initiated the idea of paper. Made presentation at ICODO 2023 conference. Wrote first draft of abstract and conclusion section. Revised whole paper critically for important intellectual content.

Funding Not applicable. No funding was received for conducting this study.

Data Availability No actual data was employed in this study; instead, theoretical constructs and artificially generated data were utilized for simulation purposes facilitating the comprehension of economic theory by readers without an economic background. The details of the artificially generated data for simulation are explicitly provided in Tables 1(1), 1(2), 2(1), 2(2), 3(1), and 3(2).

Declarations

Competing Interests Not applicable. No competing interests exist for this research. The authors have no relevant financial or non-financial interests to disclose.

The preliminary iteration of this study was presented at ICODO 2023 under the title "Modeling Intelligence Augmentation: Implications from Heterogeneous Industries and Economies." Subsequently, we have undertaken a substantial expansion of this work. It is important to affirm that this extended research paper is entirely original and is not presently undergoing evaluation or being contemplated for publication in any other journals, conferences, or.

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Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

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